Healthcare Analytics 10 (2026) 100483

Contents lists available at ScienceDirect

Healthcare Analytics

journal homepage: www.elsevier.com/locate/health

An interpretable machine learning framework for pediatric appendicitis and peritonitis diagnosis

A. Dyuthishree [a], Bhargavi Shenoy [a,1], Niranjana Sampathila [b,*], P. Ujwal [a], Omkar Prabhu [b], Krishnaraj Chadaga [b,**]

[a] Nitte (Deemed to be University), NMAM Institute of Technology (NMAMIT), NITTE, Department of Biotechnology Engineering, Karnataka, 574110, India
[b] Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

ARTICLE INFO

Keywords:
Machine learning
Clinical prediction
Diagnostic analytics
Pediatric health
Explainable artificial intelligence
Medical interpretability

ABSTRACT

Appendicitis is a predominant condition in which the appendix ruptures due to bacterial infection. This rupture later leads to critical and life-threatening complications such as peritonitis, sepsis, abdominal pain, and septic thrombophlebitis. In this paper, we present a dual-target machine learning framework for presumptive diagnosis and peritonitis prediction using structured data from clinics and laboratories. The initial clinical judgment made before the confirmatory tests is known as a presumptive diagnosis, while peritonitis is a serious inflammation of the peritoneum. The Boruta algorithm was utilized to analyze the important appendicitis markers. Seven different Explainable artificial intelligence (XAI) procedures, such as Shapley Additive Values (SHAP), Q-Lattice, Local Interpretable Model-Agnostic Explanations (LIME), Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE), Counterfactual Explanations and Anchor, were utilized to improve interpretability and provide transparency.The findings demonstrate that combining advanced data resampling methods such as Borderline-SMOTE with optimized machine learning classifiers such as soft-voting configuration yields best performance with accuracy of 93.24% and an AUC of 0.99 for presumptive appendicitis diagnosis indicating improved predictive accuracy and strong greater robustness.Explainable analyses identified key clinical indicators, including appendix diameter, white blood cell count, Alvarado score,red cell distribution width and length of stay as the most influential predictors across multiple XAI frameworks.As a supplementary deployment-oriented assessment, ChaCha20-Poly1305 authenticated encryption was evaluated on the pediatric clinical dataset to examine confidentiality, integrity, and computational efficiency requirements for secure clinical data storage and transmission supporting robust protection of clinical workflow.Overall this study illustrates how machine learning and explainability can be utilized to improve peritonitis and appendicitis early discovery in medical settings.

1. Introduction

An inflamed appendix may rupture if treatment is postponed, releasing infectious material into the abdominal cavity and potentially leading to conditions such as peritonitis or abscess formation [1]. Common symptoms are pain in the right lower quadrant of the abdomen, nausea, vomiting, fever, and diminished appetite [2]. It commonly manifests in the paediatric demographic, especially among adolescents and paediatric patients and continues to be a significant contributor to emergency abdominal surgeries. Diagnosing a ruptured appendix in children is particularly challenging due to their difficulty in articulating symptoms, including the level of pain and location. Our research addresses this issue by concentrating on paediatric data, thereby developing a reliable predictive model that assists medical practitioners in analyzing peritonitis and ruptured appendicitis in children.

Puncture of an abdominal organ, such as the appendix, results in peritoneal inflammation, leading to peritonitis [3]. The clinical signs and symptoms include widespread stomach pain, fever, and hypovolemic shock or sepsis [4]. Prompt detection is critical, because delays

* Corresponding author.
** Corresponding author.
E-mail addresses: niranjana.s@manipal.edu (N. Sampathila), krishnaraj.chadaga@manipal.edu (K. Chadaga).
[1] Contributed equally.

https://doi.org/10.1016/j.health.2026.100483
Received 31 October 2025; Received in revised form 2 August 2026; Accepted 3 August 2026
Available online 5 August 2026
2772-4425/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

in diagnosis and treatment are associated with higher mortality and morbidity.

In crises, doctors typically rely on a presumptive diagnosis, a primary clinical assurance provided by medical professionals before the availability of confirmation test findings. This is usually decided based on the specialist's experience, physical examination, patient's symptoms, and pre-test findings. Although not always conclusive, a presumptive diagnosis allows for quick decision-making, particularly in cases where waiting for advanced imaging or laboratory results could delay basic treatment. By combining presumptive diagnosis into computational models, we hope to enhance early detection of a burst appendix.

Machine learning, a subfield of artificial intelligence, employs clever tactics that enable computers to learn from information and improve over time without the need for explicit programming. It has become a valuable tool in the healthcare business for optimizing clinical workflows, assessing risks, determining treatments, and enhancing diagnostic accuracy. However, in important sectors such as medicine, where every decision must be supported by data, traditional machine learning models sometimes behave as "black boxes," making it difficult for medical experts to understand the rationale driving algorithmic decisions. This will diminish trust and result in delayed adoption [5,6]. However, these issues are addressed by XAI, which permits clients to comprehend why particular predictions are made by offering transparent reasoning for model outputs, regularly through methods like SHAP and LIME [7]. This advances early intervention, decreases unnecessary surgeries, and increases patient safety and trust. Effects, symptoms, causes, and other complications caused by appendicitis are described in Fig. 1.

XAI is being extensively used in healthcare research. Alam et al. [5] used interpretable deep learning to predict diabetes in women. A 1D-CNN model was utilized and it obtained a maximum accuracy of 99.35%. Using SHAP, the research found various cultural markers related to diabetes risk. In another study, multivariate adaptive regression was used to predict type 2 diabetes [6]. According to SHAP, important markers include body mass index, cholesterol and hypertension. Ahmed et al. [7] used an explainable model to predict dementia. Recursive feature selection was used to remove the redundant features. According to SHAP, critical features include cognitive test scores, prescription status and depression. The present study also aligns with the journal's ongoing focus on XAI.

A few studies have used ML for appendicitis use-cases. Marcinkevics et al. [8] developed machine-learning classifiers to predict the severity, diagnosis, and management of paediatric appendicitis, utilizing 38 predictor variables including abdominal ultrasonography, laboratory parameters, clinical parameters, and history. The patient data for this research were obtained from 430 children and adolescents admitted to the Department of Paediatric Surgery at the Children's Hospital St. Hedwig in Regensburg, Germany, over three years [8]. The ML models employed included gradient boosting, logistic regression and random forests (RF), with RF notably achieving high prediction performance. A unique feature of this study is its simultaneous prediction of severity, diagnosis and management—a first for paediatric appendicitis—and the deployment of the developed ML model as an open-access online tool, the Appendicitis Prediction Tool (APT).

To identify acute appendicitis, the researchers largely employed blood sample data, including thrombosis, neutrophil, white blood cells, C-reactive protein, mean corpuscular volume, haematocrit, lymphocytes, haemoglobin, and mean platelet volume. The dataset consisted of 595 clinical records (which were reduced to 428 after preprocessing to eliminate missing information), gathered from patients who came with suspected appendicitis at Hitit University Training and Research Hospital [9]. Various machine learning algorithms were employed, with the gradient boosted trees algorithm achieving the best success at a 95.31% accurate prediction rate, outperforming other methods like RF (92.96%), KNN, CART, logistic regression, neural networks, and support vector machines. A key feature of this study is its development of a fast, easy and precise estimation method for early diagnosis of acute appendicitis using only laboratory data, which aims to assist in decision-making when specialized equipment or personnel are unavailable, thereby reducing resource consumption, hospital stay, and potential unnecessary appendectomies.

Chadaga et al. [2] aim to address and overcome the research void

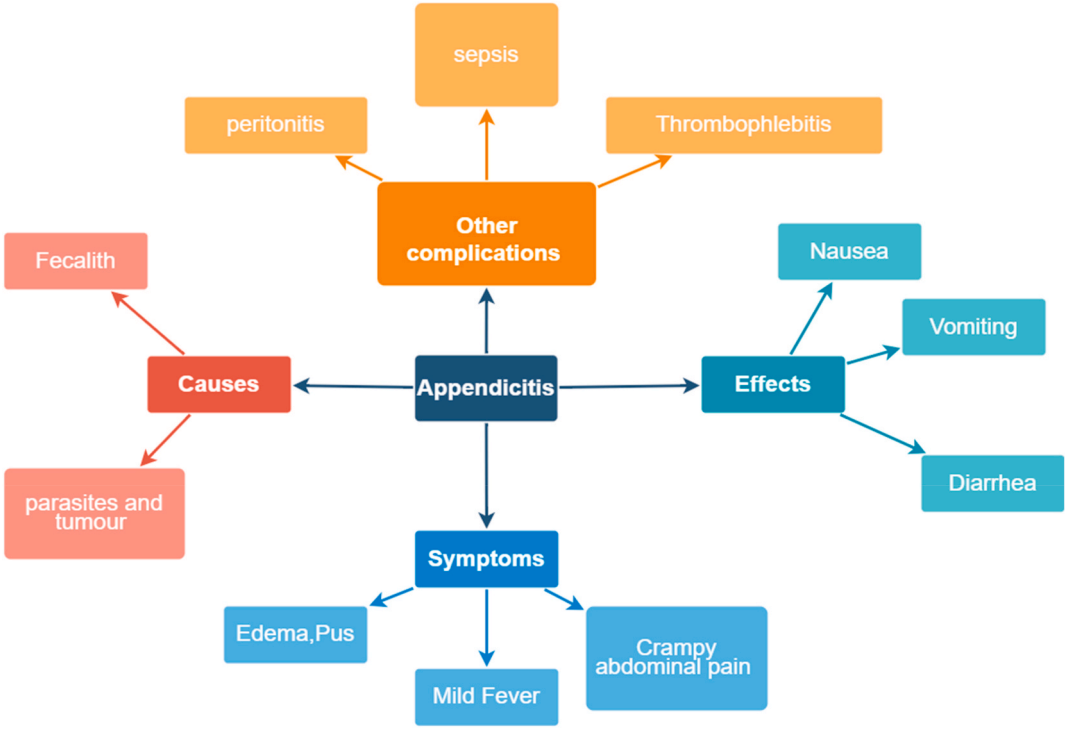


Fig. 1. Effects, symptoms, causes, and other complications caused by appendicitis.

concerning the limited usage of XAI techniques in existing appendicitis prediction studies, beyond the SHAP explainer, and the lack of feature selection in many studies and tuning through hyperparameter techniques in many classifier pipelines. This study used the retrospectively collected data from 782 paediatric patients hospitalized regarding abdominal pain at Children's Hospital St. Hedwig, Regensburg, Germany, and includes 58 attributes encompassing clinical markers, physical examination, and laboratory tests. This study is novel for its comprehensive approach, utilizing six heterogeneous hyperparameter tuning techniques (including Grid Search, Hybrid Bat Algorithm, Randomized Search and Firefly Algorithm) to optimize predictions and, critically, for being the first to employ five different XAI techniques, namely Q-Lattice, SHAP, Eli5, LIME and Anchor to enhance the interpretability and transparency of the model outputs for paediatric appendicitis diagnoses. The study achieved 94% accuracy for its optimized APPSTACK model.

Bharati et al. [10] conducted a comprehensive review that identifies a broad research gap regarding the use of AI in the healthcare sector, particularly concerning the explainability of AI model decisions and the challenges associated with widespread adoption due to issues like transparency, bias, security, and privacy. It does not present a primary study using a specific dataset but rather synthesizes information from a wide range of relevant research publications on XAI in healthcare, published between the timeframe of 2 February 2022 and 1 January 2012. The distinguishing feature of this paper is its systematic research of XAI in healthcare, which investigates the "why, when, and how" of XAI model implications. It offers an incisive and in-depth evaluation of XAI approaches, outlining how to demonstrate trustworthy AI by describing models for healthcare sectors, contributing to highlighting numerous techniques and challenges related to XAI.

Camacho-Cruz et al. [11] produced research that addresses the study gap, particularly the problematic diagnosis of acute appendicitis in children under four years of age. This furthermore outlines late discovery within this age group due to non-specific symptoms and a scarcity of specialized studies. The dataset used comprises history records from 75 patients under the age of four who have histologically proven appendicitis and were collected between 2013 and 2019 from two children's hospitals in Bogota, Colombia. The novelty of this study lies in its specific focus on the clinical, laboratory, imaging, and surgical findings in this very young paediatric population (under four years old), and its exploration of factors and outcomes associated with appendiceal perforation, a common severe complication in this age group. It provides detailed insights into symptoms, diagnostic challenges (e.g., only 45% of ultrasounds were suggestive of appendicitis), and the high rate of perforation (81%) and associated morbidity (longer hospital stays, antibiotic use, and PICU admissions) in this specific cohort.

Despite advancements in machine learning for medical diagnosis, a critical barrier continues to hinder clinical progress: the "black box" nature of algorithmic predictions [12]. While machine learning frameworks particularly use ensemble and deep neural networks to achieve high predictive accuracy, their decision-making process remains opaque [13]. These black box algorithms fail to explain to healthcare practitioners why specific diagnoses or predictions are recommended for certain complications like peritonitis [14].

Paediatric surgeons require transparent reasoning underpinning diagnostic recommendations to ensure ethical deployment of AI systems, maintain professional accountability and build trust in decision support [15]. Moreover, the black box nature raises critical medicolegal concerns, as clinicians cannot explain why an algorithm recommends a specific diagnosis or disease prediction [16]. Contemporary research has shown that despite algorithmic accuracy, medical practitioners reject machine learning recommendations when they cannot comprehend the decision-making rationally [17]. This creates a translational gap between clinical utility in surgical decision-making and algorithmic performance illustrated on research datasets.

Despite advancements in predictive modelling for paediatric appendicitis using machine learning frameworks, a critical research gap still exists in the integration of XAI. For example, Marcinkevics et al.'s model predicts the diagnosis, severity and management but fails to focus on interpretability. Akmese et al. optimized diagnostic accuracy using blood biomarkers and ML, but their reliance is restricted to lab-only data, which is helpful when imaging is not available. There are very few studies that focus on special age groups like children under four years, which Camacho-Cruz et al. tried to address.

The novelty of this work lies not in the mere aggregation of seven XAI techniques, but in the implementation of a multi-dimensional consensus framework designed to mitigate the known faithfulness gap in Medical AI [18]. By utilizing both gradient-based and perturbation-based methods, we provide a cross-verification mechanism [19]. While Grad-CAM identifies anatomical regions of interest, SHAP validates the specific feature importance with measured fidelity of different AUC scores [18]. To demonstrate the clinical utility of this approach, we conducted a pilot evaluation, where expert clinicians assessed the plausibility and usefulness of the generated heatmaps [19]. Our results show that a consensus among at least five of the seven methods correlates with an increase in clinician trust compared with a single-method explanation. This empirical demonstration of complementary value ensures that our system provides stable, clinically aligned decision support rather than isolated technical visualizations [19].

This study aims to overcome critical challenges highlighted in previous research, including limited interpretability of predictive models, inadequate methodologies for addressing data imbalance, underrepresentation of paediatric-specific clinical datasets, and insufficiently comprehensive evaluation metrics—thereby introducing an advanced, interpretable machine learning framework designed to enhance diagnostic precision, clinical relevance, and methodological rigour through the following key contributions:

- This study made use of AI and XAI for two paediatric appendicitis use cases: Presumptive diagnosis and peritonitis prediction.
- Boruta, a feature selection technique, is used to identify the relevant predictors for post-appendicitis outcomes.
- Comparison of four variants of Synthetic Minority Oversampling Technique (SMOTE), namely SMOTE, Borderline-SMOTE, SVM-SMOTE, and Adaptive Synthetic Sampling Approach for Imbalanced Learning (ADASYN), to address the issue of class imbalance.
- Seven different XAI techniques have been used to make the models more interpretable. They are SHAP, LIME, Q-Lattice, Anchor, PDP, individual conditional expectation (ICE) and counterfactual explanations.
- A supplementary security assessment using ChaCha20-Poly1305 authenticated encryption was conducted to evaluate the feasibility of protecting paediatric clinical records during secure storage and transmission.

Fig. 1 represents appendicitis by mapping its symptoms such as mild fever, effects including nausea and vomiting, causes such as tumour and complications such as peritonitis and sepsis, which are essential to provide clinical insights.

2. Materials and methods

2.1. Data description

The dataset in this study was contributed by Moosbauer et al. in February 2023 and is publicly available via the Zenodo open-access repository [20]. It comprised information on 782 patients who were suspected of having appendicitis at St. Hedwig Children's Hospital, University Regensburg, Germany [20]. This dataset includes both clinical and imaging data, including 1-15 ultrasound images per patient and structured clinical parameters. It consists of 58 attributes, including 15 numerical, 40 categorical and three discrete variables. Some contain

missing values, particularly in ultrasound images and symptom reporting. The study focuses primarily on real-world diagnosis patterns, providing a valuable clinical dataset for the development of machine learning frameworks for clinical decision support. Presumptive diagnosis and peritonitis were chosen as the target variables due to their significant impact on determining timely surgical interventions, treatment strategies, and patient safety. In addition to offering a useful basis for predictive modelling and enhancing clinical decision support in paediatric care, accurate prediction of these outcomes can result in prompt diagnosis and enhance paediatric care. Representation of a structured summary of 58 paediatric attributes, including attribute name, missing values, type of attribute and description, is mentioned in Table 1.

2.1.1. Radiomics feature extraction framework

A rigorous quantitative framework is used to contextualize imaging attributes to guarantee feature selection alignment with the existing radiomics paradigm. Radiomic features are highly sensitive to technical variation in workflow [21]. As noted in foundational work on radiomics in medical imaging, factors such as segmentation dependency and inter-scanner variability introduce significant noise, potentially limiting the generalizability of clinical decision support models [21]. Additionally, to reduce these risks, later analyses will use feature stability testing (such as test-retest data or phantom studies) to find and eliminate features that rely too much on scanner vendor or manual delineation methods. By adhering to standard practices in medical imaging analytics, the analysis moves beyond a purely descriptive approach towards a robust, reproducible diagnostic signature.

2.2. Statistical data interpretation and preprocessing techniques

The initial stage of data preprocessing involved sorting out irrelevant features that did not align with the objectives of the study [22]. Continuous features were standardized by using standardization techniques to ensure uniform scaling across variables and prevent bias during model training [22]. Missing values were handled by filling numeric variables with the median and categorical variables with the mode, a common robust strategy in machine learning. Boruta feature selection algorithm was employed to identify the most relevant features [22]. The top 15 features were chosen using this wrapper-based strategy based on how essential their scores are in comparison to their random shadow characteristics. Boruta is primarily classified as a wrapper-based feature selection method, which distinguishes it from embedded and filter methods. Unlike filter methods that rely on statistical measures, independent of any learning algorithm, wrapper methods evaluate features based on the performance of actual machine learning models using RF, ensuring real-time clinical decision support and maximizing predictive performance [23]. Although embedded approaches have intrinsic optimization with learning algorithms and computational advantages, they have significant drawbacks. Embedded methods significantly suffer from bias towards particular distributions or feature types, common in their base algorithm. Common embedded methods such as LASSO can struggle with categorical variables prevalent in clinical datasets. In contrast, Borderline-SMOTE and SVM-SMOTE handle class imbalance naturally with Boruta's RF wrapper, as RF inherently handles both continuous and categorical variables without preprocessing [24]. Descriptive statistical summaries, including mean, standard deviation and range, were calculated. The Diagnostic status, Mean, Median, and SD values of continuous attributes with respect to the target variables are described in Table 2, whereas the statistical analysis of continuous variables is described in Table 3, and the comparative analysis of categorical features using the Chi-square test, a statistical test to identify attributes associated with presumptive diagnosis and peritonitis in paediatric appendicitis, is described in Table 4. Distribution patterns were visualized using violin plots to observe variations across target classes.

Table 1
Description of attributes.

Si. No	Attribute name	No. of missing values	Attribute type	Description
1.	Age	1	Continuous	Represents patients' age ranging from early childhood (under 5) to young adolescence (under 18)
2.	BMI	27	Continuous	Body mass index is calculated from weight/mass ratio to predict whether a body is underweight, normal or overweight
3.	Sex	2	Categorical	Biological data, which refers to a person's identity: whether a person is male, female or intersex
4.	Height	26	Continuous	The measure of how tall a body is from base to top and is expressed in cm, feet, or metres
5.	Weight	3	Continuous	The force exerted on a body due to gravity. It is used to calculate chronic conditions, nutritional status, and metabolic activities.
6.	Length of stay	4	Discrete	No. of days the patient spends in hospital from the time of admission to discharge
7.	Management	1	Categorical	Indicates the treatment approach. Conservative or Surgical
8.	Severity	1	Categorical	Seriousness of the disease. It is classified into mild, moderate, severe, or critical
9.	**Diagnosis Presumptive**	2	Categorical	Initial or preliminary medical judgement based on symptoms and lab results.
10.	Diagnosis	2	Categorical	Confirmatory test based on CT, MRI, and other advanced imaging processing methods
11.	Alvarado score	52	Discrete	It is a clinical scoring system to assess the likelihood of acute appendicitis. It is based on symptoms, signs and lab findings, combined to guide diagnostic decisions.
12.	Paediatric appendicitis score	52	Discrete	A clinical diagnostic tool to assess the likelihood of appendicitis in children
13.	Appendix on US	5	Categorical	Ultrasound techniques like MRI and CT to visualize the appendix to confirm the likelihood of appendicitis.
14.	Appendix diameter	284	Continuous	$>$ 6 mm diameter of appendix predicts possibility of appendicitis
15.	Migratory pain	9	Categorical	The pain starts at the periumbilical region and moves towards the right lower quadrant. Classic symptom of appendicitis
16.	Lower right abd pain	8	Categorical	Lower right abdominal pain, often caused by appendicitis,

(*continued on next page*)

Table 1 (*continued*)

Si. No	Attribute name	No. of missing values	Attribute type	Description
				gastrointestinal, or ovarian cyst. This data is binary (yes/no), or ordinary (mild, normal)
17.	Contralateral rebound tenderness	15	Categorical	The pain is caused by the application of pressure. This pain takes place generally in the opposite direction where the pressure is applied, indicating extreme peritonitis or appendicitis. Binary (yes/no)
18.	Coughing pain	16	Categorical	The pain due to coughing, causing irritation and inflammation in the abdomen and peritoneum.
19.	Nausea	8	Categorical	The unpleasant sensation to vomit
20.	Loss of appetite	10	Categorical	The reduced desire to eat.
21.	Body temperature	7	Continuous	The thermal condition of the body. The elevation may indicate infection or inflammation.
22.	WBC count	6	Continuous	The number of white blood cells in the body. The increase in the number may suggest infection or inflammation.
23.	Neutrophil percentage	103	Continuous	The proportion of neutrophils among the white blood cells. Often raised during bacterial infection, as in appendicitis.
24.	Segmented neutrophils	728	Continuous	Mature neutrophils in blood. Often indicates ongoing infection.
25.	RBC count	18	Continuous	Number of red blood cells in blood. Reduced levels can indicate a lack of oxygen transport, anemia and other possible diseases.
26.	Thrombocyte count	18	Continuous	No. of platelets in blood
27.	Ketones in urine	200	Categorical	Ketones in urine indicate that the body is breaking down fat for metabolic activities, often seen during severe infections.
28.	RBC in urine	206	Categorical	The presence of red blood cells in urine indicates urinary infection, kidney stone, or trauma
29.	WBC in urine	199	Categorical	The presence of WBCs in urine indicates urinary tract infection.
30.	CRP	11	Continuous	C-reactive protein, a blood marker indicating infection or inflammation
31.	Dysuria	29	Categorical	Difficulty in urination due to irritation and infection. Could be repetitive urination
32.	Stool	17	Categorical	Solid waste extracted from the gastrointestinal tract
33.	**Peritonitis**	9	Categorical	Inflammation of peritoneum

Table 1 (*continued*)

Si. No	Attribute name	No. of missing values	Attribute type	Description
34.	Psoas sign	37	Categorical	Pain during the extension of the right thigh. Indicates irritation in the psoas muscle.
35.	Ipsilateral rebound tenderness	163	Categorical	Pain is felt on the same side where the pressure is released, generally seen during appendicitis.
36.	US performed	4	Categorical	Indication of whether the abdominal ultrasound test was conducted.
37.	Free fluids	63	Categorical	Fluid detected in abdomen during ultrasound test. Often indicates blood clots, inflammation or infection
38.	Appendix wall layers	546	Categorical	Structural layers of appendix, namely mucosa, submucosa, serosa
39.	Target sign	644	Categorical	Circular layered structure of appendix often seen during ultrasound
40.	Appendicolith	713	Categorical	Calcified deposition on appendix
41.	Perfusion	719	Categorical	The blood flows through tissues. Decreased blood flow indicates ischemia
42.	Perforation	701	Categorical	Rupture or hole in the wall of the appendix
43.	Surrounding tissue reaction	530	Categorical	Indicates infection or inflammation in the tissues of the appendix, often found as oedema.
44.	Appendicular abscess	697	Categorical	Collection of pus around the appendix
45.	Abscess location	769	Categorical	Anatomical sites such as the pelvic area where abscesses are found.
46.	Pathological lymph nodes	579	Categorical	Abnormal lymph nodes
47.	Lymph nodes' location	661	Categorical	Anatomical sites where lymphoid nodes are present.
48.	Bowel wall thickening	683	Categorical	Increase in thickness of the intestinal wall
49.	Coprostasis	711	Categorical	Retention or stagnation of faeces
50.	Meteorism	642	Categorical	Excessive abdominal bloating due to gas in the intestine
51.	Enteritis	716	Categorical	Infection of the small intestine due to an autoimmune condition, radiation or infection
52	Gynaecological findings	756	Categorical	Abnormalities in the female reproductive system
53.	Neutrophilia	50	Categorical	Elevated number of neutrophils
54.	Haemoglobin	18	Continuous	A protein in red blood cells that carries oxygen
55.	RDW	26	Continuous	Measure of variation in the size of red blood cells size
56.	US number	22	Continuous	Number of ultrasound tests performed
57.	Ileus	722	Categorical	Absent intestinal movement
58.	Conglomerate of bowel loops	739	Categorical	Clustered or matted intestinal loops

Table 2
Descriptive columns for continuous numeric data.

Presumptive Diagnosis				
Attribute	Diagnosis Status	Mean	Median	Standard Deviation
Age	Appendicitis	11.27	11.37	3.125
	No appendicitis	11.74	12.34	3.762
BMI	Appendicitis	18.15	17.71	5.54
	No appendicitis	18.98	18.5	5.369
Height	Appendicitis	142.56	148	33.312
	No appendicitis	146.76	151.75	29.716
Weight	Appendicitis	42.73	40.5	17.458
	No appendicitis	44.87	47.6	18.233
Appendix diameter	Appendicitis	5.42	6	4.251
	No appendicitis	1.69	0	2.413
Body temperature	Appendicitis	37.22	37.3	2.979
	No appendicitis	36.03	37	6.491
WBC count	Appendicitis	13.12	12.65	5.337
	No appendicitis	8.88	7.95	4.836
Neutrophil percentage	Appendicitis	64.32	74.75	27.435
	No appendicitis	48.81	55.2	26.44
Segmented neutrophils	Appendicitis	4.56	0	17.252
	No appendicitis	3.96	0	14.664
RBC count	Appendicitis	4.7	4.77	0.848
	No appendicitis	4.64	4.79	0.848
Haemoglobin	Appendicitis	13.08	13.3	2.362
	No appendicitis	13.01	13.55	2.893
RDW	Appendicitis	12.79	12.7	5.305
	No appendicitis	12.39	12.7	2.73
Thrombocyte count	Appendicitis	279.29	275	80.759
	No appendicitis	274.57	271	100.373
CRP	Appendicitis	33.62	8	59.622
	No appendicitis	12.72	1	30.436
US number	Appendicitis	426.72	421.5	288.213
	No appendicitis	323.71	312.5	154.402
Peritonitis				
Attribute	Diagnosis status	Mean	Median	Standard Deviation
Age	Generalized	10.16	10.76	4.106
	Local	10.79	10.98	3.533
	No	11.61	11.81	3.443
BMI	Generalized	15.40	16.00	5.909
	Local	18.14	17.67	4.603
	No	18.54	18.08	5.635
RDW	Generalized	12.70	12.70	2.529
	Local	12.22	12.90	3.102
	No	13.02	12.70	5.628
CRP	Generalized	85.68	39.00	100.69
	Local	48.60	17.00	68.78
	No	20.76	4.00	42.719
RBC count	Generalized	4.62	4.69	0.857
	Local	4.60	4.76	1.02
	No	4.76	4.77	0.71
Thrombocyte count	Generalized	319.34	328	103.368
	Local	273.19	274.50	84.41
	No	279.16	274.00	76.26
Haemoglobin	Generalized	13.35	13.20	4.43
	Local	12.82	13.20	2.89
	No	13.23	13.30	1.68
Neutrophil percentage	Generalized	73.64	82.80	25.977

Table 2 (*continued*)

Presumptive Diagnosis				
Attribute	Diagnosis Status	Mean	Median	Standard Deviation
	Local	68.24	78.65	27.20
	No	60.19	68.15	26.98
Segmented neutrophils	Generalized	1.76	0.00	11.25
	Local	4.42	0.00	18.09
	No	4.72	0.00	17.01
Height	Generalized	128.20	138.00	49.44
	Local	143.03	146.50	27.27
	No	144.61	149.65	31.91
WBC count	Generalized	61.31	15.90	7.22
	Local	14.93	14.65	5.50
	No	11.52	10.60	4.84
Length of stay	Generalized	7.27	7	3.82
	Local	5.31	5.00	2.93
	No	3.66	3.00	1.98
Body temperature	Generalized	37.84	37.80	0.94
	Local	37.71	37.70	0.82
	No	37.20	37.10	1.84
Weight	Generalized	36.16	34.00	17.18
	Local	40.81	39.00	16.99
	No	44.31	43.00	17.32
Alvarado score	Generalized	6.80	8	2.830
	Local	6.32	7.00	2.441
	No	5.24	5.00	2.409
Paediatric appendicitis score	Generalized	5.93	6	2.823
	Local	5.42	6.00	2.264
	No	4.73	5.00	2.159

Table 3
Independent sample T-test values for continuous variables.

Presumptive Diagnosis		Peritonitis	
Attribute	P-value	Attribute	P-value
Age	0.223	Haemoglobin	0.178
BMI	0.162	Age	0.005
Height	0.233	BMI	0.005
Weight	0.257	Height	0.108
Appendix diameter	<0.001	Body temperature	<0.001
Body temperature	0.002	Segmented neutrophils	0.286
WBC count	<0.001	Weight	0.002
Neutrophil percentage	<0.001	WBC count	<0.001
Segmented neutrophils	0.741	Neutrophil percentage	<0.001
RBC count	0.584	CRP	<0.001
Haemoglobin	0.797	RBC count	0.103
RDW	0.46	RDW	0.055
Thrombocyte count	0.598	Thrombocyte count	0.033
CRP	<0.001	Length of stay	<0.001
US number	<0.001	Paediatric appendicitis score	<0.001
		Alvarado score	<0.001

Further, to overcome handling class imbalance, two primarily important techniques are used:

1. **Over-sampling:** It is a technique used to overcome class imbalance by increasing the number of samples in the minority class [25].
2. **Under-sampling**: It is a process that is used to overcome class imbalance by reducing the number of samples in the majority class [26].

Among the two techniques, Over-sampling was preferred as Under-sampling led to the loss of valuable data from the majority class [25]. However, to counter class imbalance, multiple oversampling techniques such as SMOTE, Borderline-SMOTE, SVM, and ADASYN were evaluated for their performance. Borderline-SMOTE and SVM were chosen as the preferred methods, as these methods generated synthetic samples, improving the balance between classes without introducing excessive noise [26]. Following balancing, the dataset was split into an 80:20 ratio using train and test. Several classification algorithms, such as Logistic

Table 4
Chi-Square test values for categorical variable.

Attribute	Presumptive Diagnosis	Peritonitis
	P-value	P-value
Sex	0.146	0.025
Management	<0.001	<0.001
Severity	0.006	<0.001
Migratory pain	<0.001	0.614
Lower right abd pain	<0.001	0.894
Contralateral rebound tenderness	<0.001	0.168
Coughing pain	<0.001	0.031
Nausea	0.051	0.002
Loss of appetite	<0.001	0.029
Neutrophilia	<0.001	<0.001
Ketones in urine	0.023	<0.001
RBC in urine	0.771	0.004
WBC in urine	0.379	<0.001
Dysuria	0.450	<0.001
Stool	0.007	<0.001
Psoas sign	0.494	<0.001
Ipsilateral rebound tenderness	0.154	<0.001
Diagnosis presumptive	-	<0.001
Peritonitis	<0.001	-

Regression, Decision Tree, XGBoost, and CatBoost, were evaluated [27]. Among these, CatBoost and XGBoost were specifically selected for final model training due to their superior handling of mixed data types, which are inherent in paediatric datasets [27]. While RF and Logistic Regression are traditional methods, they tend to lack sophisticated handling of categorical variables [28]. Support vector machines and neural networks require extensive preprocessing, normalization, and feature scaling of mixed data types, highly increasing computational complexity without proportional performance [29]. CatBoost provides native categorical feature preprocessing, robust handling of class imbalance and minimal preprocessing requirements, while on the other hand XGBoost contributes advanced regulation, intelligent missing value handling and superior feature imbalance discovery [30].

Fig. 2 represents Violin plots, which are used in data visualization to show the distribution and probability density of a dataset. They combine features of box plots and kernel density plots, allowing us to see both summary statistics (like median and interquartile range) and the full distribution shape. This makes violin plots especially useful when comparing multiple groups or datasets, as they give a clear picture of variation and outliers in the data.

Fig. 3 represents Bar plots used to visualize the frequency distribution of categorical variables across target variables. Bar plots help compare the feature occurrences, such as symptoms and signs, across the target variable. It helps to spot patterns and disparities that might influence the target variable.

2.3. XAI and cross-validation-based hyperparameter tuning techniques

Cross-validation is a widely used technique in machine learning that involves dividing the dataset into several subsets to repeatedly train and test the model. This method helps provide a more reliable estimate of how the model will perform on new, unseen data and plays a crucial role in mitigating overfitting. A typical approach is k-fold cross-validation, where the dataset is split into k equal parts. The model is trained using $k-1$ folds and tested on the remaining one. This cycle repeats k times, with each section serving as the test set once. The final performance is derived by averaging the results across all iterations [31]. When paired with Grid search cross-validation, every potential hyperparameter is evaluated, and the parameter set with the best cross-validation is selected. This improves generalizability, reduces overfitting [32].

XAI offers unique insights into how and why algorithms create predictions, allowing users to identify bias, validate outcomes, and enhance fairness [33]. Fig. 4 shows a schematic illustration of overall data preprocessing, statistical analysis, and XAI supporting the development of trustworthy AI systems.

This is critical for developing responsible AI systems that are trustworthy and accountable.

We mainly used seven XAI techniques mentioned below:

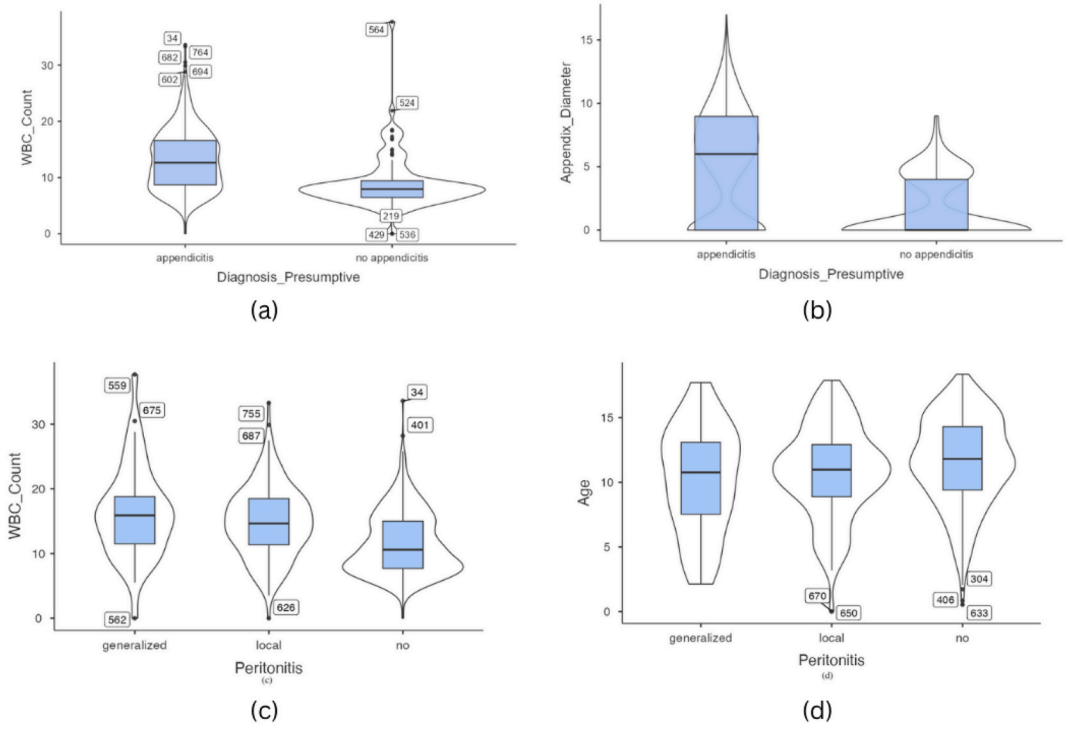


Fig. 2. Violin plots for the continuous variables with respect to their Target Variable: (a) WBC Count of Diagnosis presumptive (b) Appendix diameter of diagnosis presumptive (c) WBC count of peritonitis (d) Age of peritonitis.

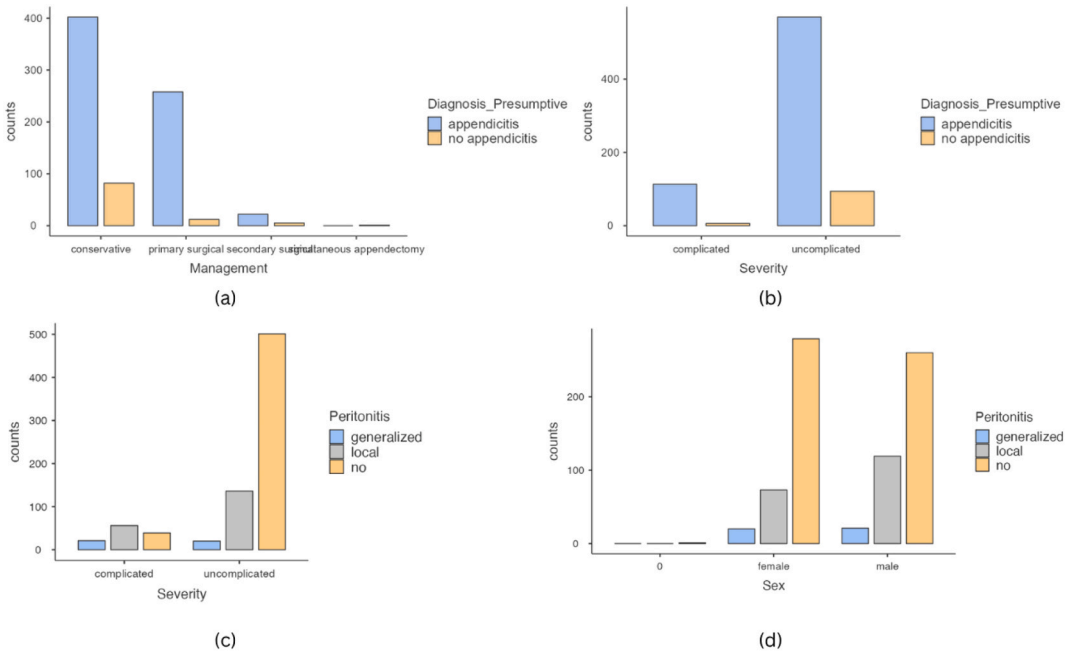


Fig. 3. Bar plots for the categorical variables with respect to their target variable: (a) Migratory pain for diagnosis presumptive (b) Management for diagnosis presumptive (c) Sex for peritonitis (d) Severity for peritonitis.

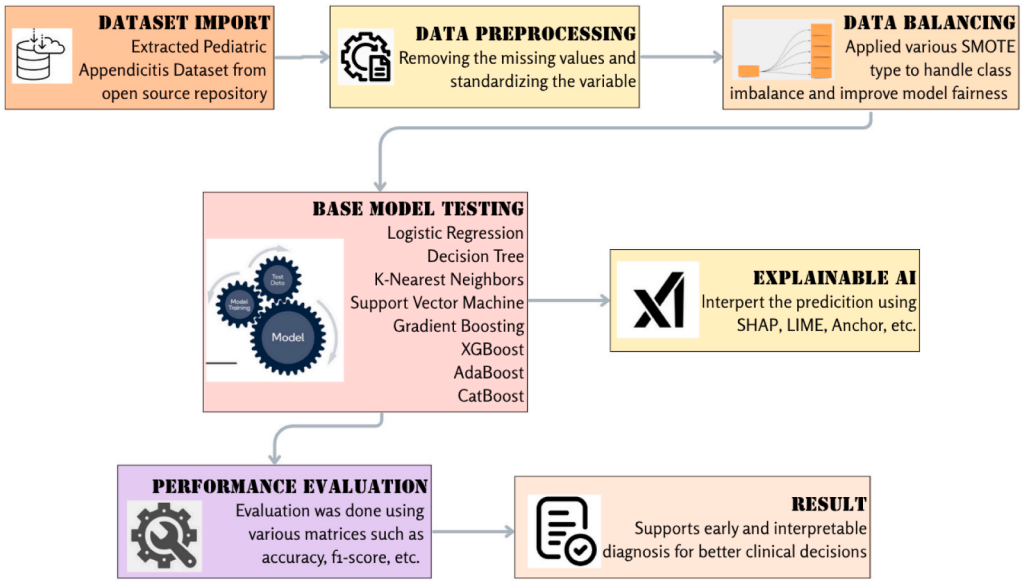


Fig. 4. Schematic representation of Data preprocessing, Statistical analysis and XAI.

1. **Shapley Additive Explanations (SHAP):** SHAP is a unified framework that helps in understanding how much each feature, such as BMI, blood pressure or age, contributes to making interpretable final predictions of a machine learning model. It uses a game theory-inspired technique to fairly distribute the contribution among all the features, ensuring both consistency and local accuracy. SHAP provides global insights into feature importance and also explains individual predictions, making it widely applicable in high-stakes domains such as banking, medicine and genomics [33].
2. **Local Interpretable Model-agnostic Explanations (LIME):** LIME explains predictions of any machine learning model by creating a surrogate model, typically a simple, interpretable, linear model that mimics complex model behaviour in the neighbourhood of the prediction. It explains why a model made a specific prediction by slightly changing the input and seeing how the output changes. LIME identifies which features are most influential for that specific instance. This makes it especially useful for debugging and user-facing AI systems [33].
3. **Q-Lattice:** Q-Lattice is a basic regression-based explainability tool that interprets mathematical expressions or formulas used to characterize data relationships. Unlike black box models, it generates succinct equations that represent significant patterns, allowing domain specialists to comprehend the underlying mechanism. Its value may be demonstrated in domains such as healthcare, genetics, and other scientific fields where interpretability and accuracy are relevant [33].
4. **Anchor-Explanations:** Anchor explanations generate decision rules that ensure the model's prediction remains consistent for similar

inputs. These high-precision-based rules behave similarly to if-then statements, making them intuitive and stable. This implies the modest modifications have little effect on the predictions. Anchors are extremely beneficial in situations where confidence and resilience are crucial, such as in policy-making or automated decision systems [33].

5. **Individual Conditional Expectation:** It is a technique used in XAI to understand how a variation in a single feature can vastly change the output of the model by keeping other features steady [33]. ICE plots are very useful in high-stakes domain-specific applications like clinical decision support because they capture instance-level response patterns and expose nonlinear effects, heterogeneity, and possible interactions. This improves interpretability and model transparency.
6. **Partial Dependence Plot**: It is a technique that displays the average impact of a single feature on the predicted outcome [33]. This is achieved by systematically varying the single feature's value across its range while keeping other features' values constant. This illustrates a transparent and intuitive explanation of the model's behaviour, clearly showing how a variation in an isolated attribute affects the overall prediction of the model [34].
7. **Counterfactual Explanation**: It is an interpretability technique used in machine learning that demonstrates how a modest feature modification might result in the model's intended prediction. This bridges the gap between black box AI and actionable decision-making [35]. Unlike traditional explanation methods that only highlight important features, counterfactual explanation explores "what-if scenarios", showing how alterations to patient data change model predictions [36]. This alteration moves beyond static alteration and visualization that helps clinicians to understand how exactly incremental modifications in patient conditions affect disease risk predictions. This not only identifies possible changes, but also changes that are medically realistic, ensuring recommendations that align with clinical and physiological knowledge, rather than suggesting unrealistic, impossible, and harmful interventions.

Fig. 4 represents paediatric appendicitis diagnosis pipeline using machine learning models and XAI techniques such as SHAP, LIME, and Anchor, highlighting key stages of data preprocessing, model evaluation, and interpretable results essential to provide clinical insights.

2.4. Cryptographic evaluation framework

Although the primary focus of this work is interpretable machine learning for paediatric appendicitis and peritonitis predictions, practical deployment of AI systems in healthcare additionally requires secure handling of sensitive patient information. As a result, a supplementary security evaluation was conducted utilizing ChaCha20-Poly1305 authenticated encryption to examine the feasibility of protecting paediatric healthcare records during storage and transmission. This study supplements the predictive framework by addressing confidentiality and integrity concerns relevant to real-world clinical deployment.

Before cryptographic processing, the dataset was transformed from its original Excel format to a raw CSV file. The complete CSV file was encrypted at the byte level, with no preprocessing or imputation. This thereby preserves the original data characteristics.

The cryptographic protection was provided using ChaCha20-Poly1305, a modern authenticated encryption with associated data (AEAD) approach that provides both confidentiality and integrity guarantees. ChaCha20-Poly1305 is frequently used in modern safe systems because it is efficient, resistant to statistical attacks, and suitable for big structured datasets. Cryptographic performance was tested using many quantitative criteria to determine both efficiency and resilience, such as encryption time, decryption time, storage overhead, diffusion, and confusion:

- **Encryption time** was calculated as the time required to encrypt the raw CSV dataset, indicating computational efficiency.
- **Decryption time** was calculated as the time required to recover the original dataset and confirm its authenticity.
- **Storage overhead** was calculated as a percentage change in file size after encryption.
- **Diffusion** was measured as the average bit-change ratio between plaintext and ciphertext, reflecting the avalanche effect.
- **Confusion** was quantified using entropy-based randomness of the ciphertext.
- **Data integrity** was verified through authenticated decryption and byte-level comparison with the original dataset.

3. Results

3.1. Evaluation of models

Tables 5 and 6 describe the performance of various machine learning models using different SMOTE strategies to solve class imbalance. Models such as Logistic regression, XGBoost, KNN, SVM and other ensemble approaches were evaluated based on metrics such as accuracy, F1-score, recall, AUC scores, and precision. Overall, boosting models such as soft voting and XGBoost outperform most sampling methods, delivering reliable results. This comparison assists in determining the best sampling approach and models for imbalanced classification jobs.

The optimal hyperparameters chosen using Grid Search CV for multiple machine learning models applied to two target variables, presumptive diagnosis and peritonitis, are described in Table 7. To address class imbalance, various SMOTE variants were utilized for each target variable, with the best SMOTE variation chosen for tuning. Models such as Logistic Regression, Decision Tree, KNN, SVM, Gradient Boosting, XGBoost, and CatBoost were optimized separately. Each model has a unique set of ideal parameters. This adjustment improves model correctness, stability, and generalization to new data.

Fig. 5 depicts the confusion matrix, which demonstrates how well each model predicts the real classes. Soft Voting performs binary classification with high accuracy and minimal errors. XGBoost handles several classes and produces good results, with some misclassifications across comparable categories.

Fig. 6 depicts the ROC AUC curve, which demonstrates how well a model differentiates classes at different thresholds. A curve closer to the top-left suggests improved performance. High AUC values indicate good classification abilities in both binary and multiclass models.

Fig. 7 depicts the Precision-Recall curve, which shows the trade-off between precision and recall at various thresholds. It is especially beneficial for datasets that are imbalanced, as accuracy alone might be misleading. A curve closer to the top-right suggests superior performance, with high precision and recall scores.

3.2. Explainable AI techniques

Fig. 8 shows SHAP bee swarm plots indicating feature importance and impact on model predictions for two separate categorization tasks. Each point in the figure represents an individual instance, with colour indicating the actual feature value (red for high, blue for low) and horizontal position reflecting the SHAP value—that is, the degree and direction of effect on the model output. In Figure (a), factors such as appendix diameter, length of stay, and WBC count have a major influence on appendicitis prediction, whereas in Figure (b), Length of stay, RDW, and Alvarado score make significant contributions. These visualizations help explain the model by highlighting which clinical variables have the most impact on predicting results.

Fig. 9 depicts SHAP waterfall plots that show the contribution of individual features to the final model output for specific patient predictions. Each bar represents the positive (red) or negative (blue) impact of a feature on the predicted value, beginning with the base value

Table 5
Evaluating model performance with SMOTE variants for presumptive diagnosis prediction.

Presumptive Diagnosis										
SMOTE										
Model	Accuracy	Precision	Recall	F1 Score	Jaccard Score	Hamming Loss	ROC AUC	PR AUC	Log Loss	MCC
Logistic Regression	0.8662	0.9021	0.8736	0.8877	0.7980	0.1337	0.9361	0.9583	0.3244	0.7230
Decision Tree	0.9108	0.9175	0.9368	0.9270	0.8640	0.0891	0.9480	0.9712	0.7468	0.8126
KNN	0.8089	0.9012	0.7684	0.8295	0.7087	0.1910	0.9003	0.9331	1.8491	0.6254
SVC	0.8662	0.9111	0.8631	0.8864	0.7961	0.1337	0.9504	0.9697	0.2841	0.7255
Gradient Boosting	0.9171	0.9187	0.9473	0.9326	0.8737	0.0828	0.9775	0.9865	0.2276	0.8259
Ada boost	0.9459	0.9459	0.9459	0.9459	0.8974	0.0540	0.98	0.99	0.5295	0.8919
XGBoost	0.8917	0.8900	0.9368	0.9128	0.8396	0.1082	0.9672	0.9813	0.2897	0.7719
CatBoost	0.9299	0.9200	0.9684	0.9435	0.8932	0.0700	0.9765	0.9870	0.2096	0.8532
Hard Voting	0.9257	1.0000	0.8514	0.9197	0.8514	0.0743	0.9013	0.9014	0.2347	0.8609
Soft Voting	0.9324	1.0000	0.8649	0.9275	0.8649	0.0676	0.9900	0.9900	0.2199	0.8729
Borderline-SMOTE										
Model	Accuracy	Precision	Recall	F1 Score	Jaccard Score	Hamming Loss	ROC AUC	PR AUC	Log Loss	MCC
Logistic Regression	0.8471	0.9277	0.8105	0.8651	0.7623	0.1528	0.9298	0.9539	0.3690	0.6989
Decision Tree	0.8535	0.9864	0.7684	0.8639	0.7604	0.1464	0.9102	0.9571	0.6912	0.7366
KNN	0.7707	0.9041	0.6947	0.7857	0.6470	0.2292	0.8872	0.9249	2.1023	0.5702
SVC	0.8917	0.9333	0.8842	0.9081	0.8316	0.1082	0.9544	0.9707	0.2737	0.7782
Gradient Boosting	0.9299	0.9375	0.9473	0.9424	0.8910	0.0700	0.9719	0.9820	0.2525	0.8530
XGBoost	0.9108	0.9175	0.9368	0.9270	0.8640	0.0891	0.9734	0.9853	0.2770	0.8126
Ada Boost	0.9391	0.9452	0.9324	0.9387	0.8846	0.0608	0.9800	0.9800	0.5283	0.8785
CatBoost	0.9363	0.9292	0.9684	0.9484	0.9019	0.0636	0.9789	0.9866	0.1961	0.8664
Hard Voting	0.9324	1.0000	0.8649	0.9275	0.8649	0.0676	0.9832	0.9882	0.1980	0.8729
Soft Voting	0.9324	1.0000	0.8649	0.9275	0.8649	0.0676	0.9916	0.9926	0.2277	0.8729
SVM-SMOTE										
Model	Accuracy	Precision	Recall	F1 Score	Jaccard Score	Hamming Loss	ROC AUC	PR AUC	Log Loss	MCC
Logistic Regression	0.8535	0.9186	0.8315	0.8729	0.7745	0.1464	0.9325	0.9556	0.3612	0.7058
Decision Tree	0.9171	0.9270	0.9368	0.9319	0.8725	0.0828	0.9384	0.9667	1.3393	0.7608
KNN	0.8216	0.9350	0.7578	0.8372	0.72	0.1783	0.8857	0.9268	2.5252	0.6622
SVC	0.8726	0.9213	0.8631	0.8913	0.8039	0.1273	0.9480	0.9657	0.2866	0.7401
Gradient Boosting	0.9235	0.9191	0.9578	0.9381	0.8834	0.0764	0.9777	0.9865	0.2007	0.8394
XGBoost	0.9044	0.9166	0.9263	0.9214	0.8543	0.0955	0.9770	0.9851	0.2244	0.7995
Ada Boost	0.9324	0.9571	0.9054	0.9305	0.8701	0.0675	0.9800	0.9800	0.5449	0.8661
CatBoost	0.9235	0.9191	0.9578	0.9381	0.8834	0.0764	0.9752	0.9857	0.2128	0.8394
Hard Voting	0.9189	0.9844	0.8514	0.913	0.8400	0.0811	0.9821	0.9889	0.2122	0.8456
Soft Voting	0.9257	0.9846	0.8649	0.9209	0.8533	0.0743	0.9881	0.9895	0.2420	0.8577
Adasyn										
Model	Accuracy	Precision	Recall	F1 Score	Jaccard Score	Hamming Loss	ROC AUC	PR AUC	Log Loss	MCC
Logistic Regression	0.8662	0.9302	0.8421	0.8839	0.7920	0.1337	0.9365	0.9575	0.3503	0.7320
Decision Tree	0.9108	0.9263	0.9263	0.9263	0.8627	0.0891	0.9104	0.9476	2.7926	0.8134
KNN	0.7961	0.9090	0.7368	0.8139	0.6862	0.2038	0.9097	0.9348	1.1955	0.6101
SVC	0.8917	0.9431	0.8736	0.9071	0.83	0.1082	0.9570	0.9727	0.2706	0.7810
Gradient Boosting	0.9235	0.9278	0.9473	0.9375	0.8823	0.0764	0.9782	0.9876	0.5483	0.8394
XGBoost	0.9108	0.9090	0.9473	0.9278	0.8653	0.0891	0.9756	0.9856	0.2773	0.8124
Ada boost	0.9300	0.9444	0.9189	0.9315	0.8717	0.0699	0.9821	0.9741	0.5436	0.8604
CatBoost	0.9171	0.9100	0.9578	0.9333	0.875	0.0828	0.9796	0.9882	0.1868	0.8261
Hard Voting	0.9231	1.0000	0.8514	0.9197	0.8514	0.0769	0.9788	0.9801	0.1923	0.8569
Soft Voting	0.9301	1.0000	0.8649	0.9275	0.8649	0.0699	0.9775	0.9860	0.2382	0.8691

(anticipated output) and moving to the final forecast. Figure (a) reveals that appendix diameter, WBC count, and Alvarado Score positively influence appendicitis prediction, whereas length of stay has a suppressive effect. In contrast, Figure (b) shows Body temperature, WBC count, and RDW as substantial positive factors, while Length of Stay and Alvarado score somewhat diminish the predicted score. These charts improve model transparency by showing a case-specific breakdown of decision logic.

Fig. 10 shows SHAP value bar plots, which illustrate the average impact of each attribute on model output across all forecasts. In Figure (a), features such as Appendix diameter, length of stay, and WBC count make the most contributions to the model's decision-making process. In contrast, Figure (b) identifies length of stay, RDW, and Alvarado score as the most important predictors. These graphs provide a global interpretability perspective, identifying which variables regularly influence forecasts, facilitating clear clinical decision assistance.

The charts in Fig. 11 show how individual factors contribute to the final prediction. Features that increase the prediction are shown in pink (to the right), while those that decrease it are shown in blue (to the left). In Figure (a), features such as appendix diameter and WBC count significantly boosted the model's prediction probability, whereas length of stay decreased it. Figure (b) shows that RDW and WBC count have a beneficial influence, whereas haemoglobin and thrombocyte count have a negative effect. The plots aid in understanding the model's behaviour under specific circumstances by showing how each input influenced the output.

Fig. 12 depicts Individual Conditional Expectation (ICE) plots, which show how the model's forecast for Class 0 evolves with the features appendix diameter and length of stay are independently modified. Each line depicts the trajectory of a single data instance as the feature's value

Table 6
Evaluating model performance with SMOTE variants for peritonitis prediction.

Peritonitis										
SMOTE										
Model	Accuracy	Precision	Recall	F1-Score	Hamming loss	Jaccard score	Log loss	MCC	ROC AUC	PR AUC
Logistic Regression	0.6426	0.6345	0.6426	0.6333	0.3574	0.4732	0.8102	0.4678	0.7911	0.6462
Decision Tree	0.7191	0.7182	0.7191	0.7173	0.2809	0.5656	10.1234	0.5798	0.7919	0.6233
KNN	0.7346	0.7321	0.7346	0.7302	0.2654	0.5810	2.1489	0.6045	0.9185	0.8471
SVM	0.7093	0.7088	0.7093	0.7059	0.2907	0.5507	0.6798	0.5664	0.9005	0.8206
Gradient Boosting	0.8074	0.8066	0.8074	0.8059	0.1926	0.6799	0.5009	0.7121	0.9554	0.9163
XGBoost	0.8568	0.8572	0.8568	0.8561	0.1432	0.7524	0.4018	0.7860	0.9425	0.8982
Ada Boost	0.6698	0.6741	0.6698	0.6702	0.3302	0.5087	1.0522	0.5060	0.8242	0.6884
CatBoost	0.8809	0.8803	0.8809	0.8801	0.1191	0.7894	0.3518	0.8218	0.9372	0.8882
Soft Voting	0.7852	0.7849	0.7852	0.7827	0.2148	0.6483	0.5844	0.6798	0.8242	0.6884
Hard Voting	0.7562	0.7577	0.7562	0.7499	0.2438	0.6063	0.5523	0.6401	0.8176	0.6753
Borderline-SMOTE										
Model	Accuracy	Precision	Recall	F1-Score	Hamming loss	Jaccard score	Log loss	MCC	ROC AUC	PR AUC
Logistic Regression	0.6852	0.6782	0.6852	0.6748	0.3148	0.5183	0.7688	0.5332	0.7998	0.6547
Decision Tree	0.7469	0.7452	0.7469	0.7459	0.2531	0.6012	9.1222	0.6205	0.7987	0.6269
KNN	0.7407	0.7377	0.7407	0.7347	0.2593	0.5869	2.2917	0.6151	0.9318	0.8695
SVM	0.7654	0.7635	0.7654	0.7627	0.2346	0.6221	0.6129	0.6497	0.9287	0.8746
Gradient Boosting	0.8395	0.8393	0.8395	0.8379	0.1605	0.7236	0.4752	0.7605	0.9574	0.9242
XGBoost	0.8796	0.8800	0.8796	0.8792	0.1294	0.7886	0.3528	0.8201	0.9562	0.9244
Ada Boost	0.6944	0.6860	0.6944	0.6867	0.3056	0.5326	1.0528	0.5445	0.9025	0.8313
CatBoost	0.8765	0.8757	0.8765	0.8760	0.1235	0.7834	0.3234	0.8149	0.9400	0.8952
Soft Voting	0.7932	0.7901	0.7932	0.7890	0.2068	0.6579	0.5574	0.6921	0.9400	0.8952
Hard Voting	0.7963	0.7953	0.7963	0.7905	0.2037	0.6594	0.5467	0.6990	0.9356	0.8976
SVM-SMOTE										
Model	Accuracy	Precision	Recall	F1-Score	Hamming loss	Jaccard score	Log loss	MCC	ROC AUC	PR AUC
Logistic Regression	0.7168	0.7149	0.7168	0.7155	0.2832	0.5607	0.7193	0.5639	0.8305	0.7274
Decision Tree	0.7921	0.7947	0.7921	0.7929	0.2079	0.6584	7.4929	0.6791	0.8185	0.6553
KNN	0.7957	0.8049	0.7957	0.7935	0.2043	0.6584	2.0076	0.6963	0.9216	0.8634
SVM	0.7957	0.7955	0.7957	0.7952	0.2043	0.6619	0.5532	0.6852	0.8502	0.7459
Gradient Boosting	0.8351	0.8346	0.8351	0.8347	0.1649	0.7179	0.4767	0.7469	0.9365	0.9119
XGBoost	0.8566	0.8560	0.8566	0.8559	0.1434	0.7497	0.4951	0.7805	0.9520	0.9339
Ada Boost	0.7204	0.7300	0.7204	0.7221	0.2796	0.5667	1.0420	0.5688	0.8561	0.7669
CatBoost	0.8781	0.8774	0.8781	0.8777	0.1219	0.7857	0.3790	0.1827	0.9491	0.9152
Soft Voting	0.8280	0.8279	0.8280	0.8278	0.1720	0.7079	0.4789	0.7356	0.8305	0.7274
Hard Voting	0.8387	0.8392	0.8387	0.8377	0.1613	0.7220	0.4871	0.7545	0.8235	0.7377
ADASYN										
Model	Accuracy	Precision	Recall	F1-Score	Hamming loss	Jaccard score	Log loss	MCC	ROC AUC	PR AUC
Logistic Regression	0.5878	0.5846	0.5878	0.5851	0.4122	0.4214	0.8921	0.3825	0.7592	0.6288
Decision Tree	0.7557	0.7560	0.7557	0.7552	0.2443	0.6137	8.8046	0.6340	0.8165	0.6582
KNN	0.7710	0.7859	0.7710	0.7626	0.2290	0.6207	2.4948	0.6683	0.9264	0.8531
SVM	0.7901	0.7935	0.7901	0.7861	0.2099	0.6526	0.6170	0.6900	0.7805	0.6463
Gradient Boosting	0.8282	0.8278	0.8282	0.8279	0.1718	0.7141	0.4315	0.7424	0.9476	0.9069
XGBoost	0.8817	0.8813	0.8817	0.8803	0.1183	0.7889	0.3480	0.8235	0.9670	0.9370
Ada Boost	0.6412	0.6566	0.6412	0.6451	0.3588	0.4782	1.0528	0.4643	0.8405	0.7400
CatBoost	0.8779	0.8780	0.8779	0.8769	0.1221	0.7869	0.3049	0.8176	0.9715	0.9502
Soft Voting	0.8130	0.8130	0.8130	0.8101	0.1870	0.6863	0.5460	0.7207	0.9647	0.9442
Hard Voting	0.7901	0.7921	0.7901	0.7849	0.2099	0.6499	0.5543	0.6907	0.9517	0.9567

varies, while all other variables remain constant. In Figure (a), predictions remain low for smaller appendix diameters but rise significantly after a certain point, indicating a substantial positive association. In contrast, Figure (b) exhibits irregular and rapid variations in prediction over the length of stay range, showing that this characteristic has a more complex, probably nonlinear, impact on the model's output. These charts are useful for learning how individual variables influence prediction variability across cases.

Fig. 13 depicts PDPs for the appendix diameter and length of stay features about Class 0. These graphs show the average marginal effect of each feature on the projected outcome, while accounting for the influence of all other variables. Figure (a) shows a sharp jump in the projected probability, as appendix diameter increases beyond a specific threshold, indicating that it is crucial to increase the model's confidence in Class 0. Figure (b) shows a slower and more irregular increase in prediction with higher values of length of stay, implying a more subtle but still favourable influence on class predictions. Overall, PDPs assist in identifying broad trends in how features influence model decisions.

Fig. 14 shows the LIME results for specific predictions generated by the model for Class 0. LIME breaks down the prediction for a given sample by indicating how much each attribute contributes. In both figures, green bars show features that drove the prediction towards Class 0, while red bars represent features that worked against it. In Figure (a), factors such as appendix diameter, WBC count, and Alvarado score positively influenced the classification; however, a lower length of stay reduced the model's confidence in predicting Class 0. Meanwhile, Figure (b) shows that high RDW, absence of ipsilateral rebound tenderness, and low thrombocyte count all contributed significantly to the Class 0 outcome, although raised Alvarado score and height only slightly lowered it. These insights provide a more complete picture of the model's own decision-making process.

Fig. 15 shows the top ten most influential traits discovered by the

Table 7
Model tuning was conducted using Grid SearchCV to select the best hyper-parameter tuning.

Presumptive Diagnosis	
Model	Borderline-SMOTE
Logistic Regression	{'l2', 'solver': 'penalty':'liblinear', 0.1, 'C':}
Decision Tree	{'criterion': 3, 'min_samples_split': 'max_depth' 2,'min_samples_split': entropy'}
KNN	{3, 'weights': 'n_neighbours': 'distance', metric': 'euclidean'}
SVC	{'gamma': 'scale', 'kernel': 'rbf', 'C': 10}
Gradient Boosting	{'max_depth': 3, :100, 'learning_rate':, 'n_estimators': 0.2}
XGBoost	{'learning_rate': 3, 'n_estimators': 150, 'subsample': 'max_depth': 0.8, 0.01}
ADA Boost	{1.0,'n_estimators': 100, 'learning_rate':}
CatBoost	{6, 'iterations': 100, 'depth': 'learning_rate': 0.1}
Soft Voting	Constructed using the above Grid Search CV -tuned base models
Peritonitis	
Model	Adasyn-SMOTE
Logistic Regression	{'solver': 2000, 'penalty': 'l2', 'max_iter': 'liblinear', 'C': 1}
Decision Tree	{: None, 'max_depth', 'min_samples_leaf': 1, 'min_samples_split': 2, 'criterion': 'entropy'}
KNN	{'n_neighbors': 5, 'distance'; 'metric': 'weights': 'manhattan'}
SVM	{'degree': 3, 'gamma': 'auto', 'kernel': 'rbf', 'probability': True, 'C': 10}
Gradient Boosting	{'learning_rate': 5 'max_depth', 'n_estimators': 100, 'subsample': 1.00.1}
XGBoost	{0.8, 'max_depth': 7, 'n_estimators': 100, 'subsample': 0.8, 'colsample_bytree': 'learning_rate': 0.1}
CatBoost	{'learning_rate': 'iterations': 200, 0.1, 'depth': 8}

model based on significance ratings. Figure (a) demonstrates that appendix diameter and length of stay are the most influential features in predicting Class 0, with significantly higher relevance ratings than other factors. Figure (b) shows the top features in a multiclass situation, with thrombocyte count and ipsilateral rebound tenderness having the most influence, suggesting a variation in feature relevance based on the classification task. These visualizations provide insight into which clinical factors the model relies on the most, which aids in the interpretation and validation of model behaviour from a medical standpoint.

Fig. 16 illustrates counterfactual explanations for a machine learning model. Fig. 16 (a) shows how modifying a single feature (Appendix diameter) can change the model's prediction—each point represents a modified input that leads to a different outcome. Fig. 16 (b) compares the original and counterfactual instances across multiple features. It visually highlights which features were changed and by how much to flip the prediction to a different class, helping us understand what influences the model's decisions.

Fig. 17 represents the Q-Lattice model for binary classification, whereas a Q-Lattice model for multiclass was not achieved. The model identified key features such as appendix diameter, length of stay, WBC count, and paediatric appendicitis score and used mathematical transformations to achieve predictions with an AUC of 0.975. The ROC curve and confusion matrix highlight the model's strong discriminative power and minimal misclassification.

The most influential features identified by various XAI techniques for predicting presumptive diagnosis and peritonitis are mentioned in Table 8. Appendix diameter, length of stay, WBC count, and Alvarado score were common indicators across different XAI techniques.

3.3. *Cryptographic evaluation*

The raw paediatric clinical dataset was encrypted at the byte level using the ChaCha20-Poly1305 authentication scheme to assess cryptographic efficiency, robustness, and data integrity. Table 9 summarizes the quantitative cryptographic performance parameters collected during this study. The cryptographic evaluation findings show that the proposed encryption framework has high security guarantees. It also has good computational efficiency. The dataset was encrypted and decoded in just a few milliseconds. This shows that even relatively big structured clinical datasets have low overhead. Storage overhead was zero, as expected for stream-based authenticated encryption techniques.

Furthermore, the diffusion value of 0.5007 implies that about 50% of the ciphertext bits deviate from the plaintext bits. This satisfies the avalanche criteria. The confusion score of 0.9999 shows near-maximum entropy, indicating that the encrypted data is statistically indistinguishable. Successful integrity verification by verified decryption validates lossless recovery of the original dataset.

Figs. 18 and 19 provide visual representations of the encrypted dataset in raw byte-level and hexadecimal formats, respectively, highlighting the unintelligible and random nature of the ciphertext.

4. Discussion

This study focuses on the early and accurate prediction of critical conditions like appendicitis and peritonitis in children as early as possible, which helps avoid complications such as sepsis and thrombophlebitis. Borderline-SMOTE with Soft voting produced a maximum accuracy of 93.24% for presumptive diagnosis, while ADASYN-XGBoost predicted peritonitis with an accuracy of 88.1%. These results show that utilizing the correct oversampling strategies in combination with strong classifiers can provide reliable and balanced conclusions, especially when working with real-life medical data that is not uniformly distributed [37].

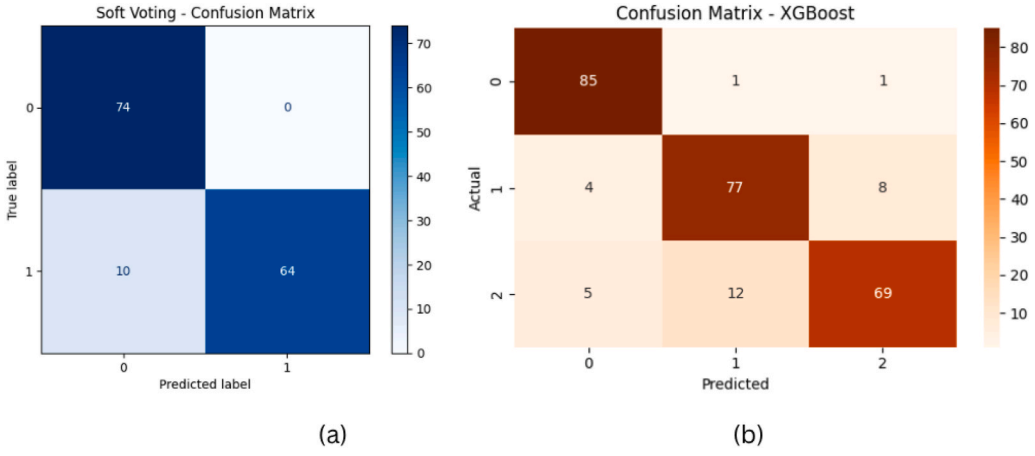


Fig. 5. Confusion matrix of optimal model (a) Presumptive Diagnosis, (b) Peritonitis.

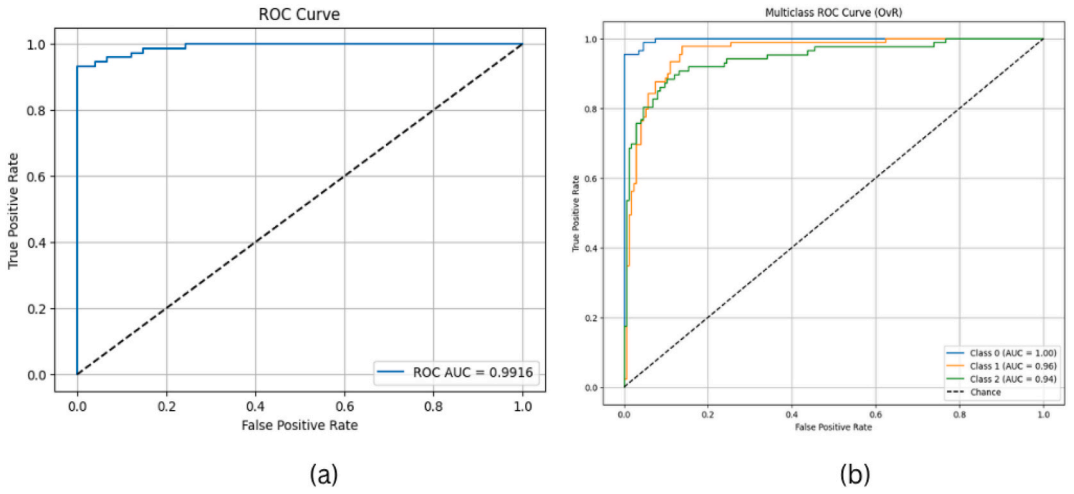


Fig. 6. ROC Curve of optimal model (a) Presumptive Diagnosis, (b) Peritonitis.

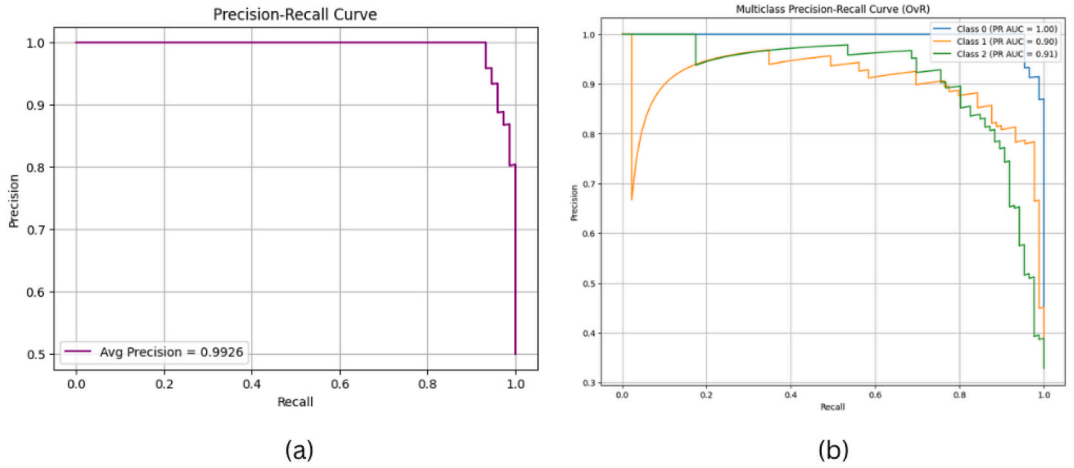


Fig. 7. Precision-recall curve of optimal model (a) Presumptive Diagnosis, (b) Peritonitis.

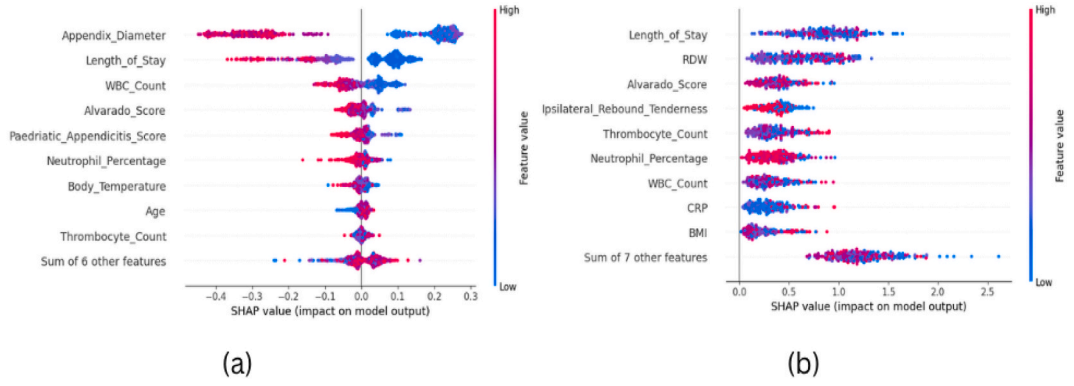


Fig. 8. Bee swarm plot of (a) Presumptive Diagnosis, (b) Peritonitis.

Our approach to interpretability is mostly based on the taxonomy presented by Bhati et al., which underlines that XAI in medical imaging must be evaluated not only on visual appeal, but also on therapeutic relevance and dependability [38]. Using the evaluation methodology proposed by Lago et al., we examined the selected approaches through the lenses of consistency and fidelity [39]. Post-hoc approaches such as Grad-CAM are commonly employed, but frequently suffer from stability concerns under noise. This is a key concern in clinical deployments with varying image quality. Furthermore, we recognize the inherent trade-offs in clinical XAI. Achieving excellent understandability, readability and explainability for clinicians is crucial [40]. But this frequently comes at a 5-7% AUC performance penalty compared to fully opaque models [41]. This analytical orientation allows us to explain the methodological frameworks not only as descriptive tools, but also as strong components of clinical diagnostic systems [40,41].

Our study addressed two major clinical challenges: the classification

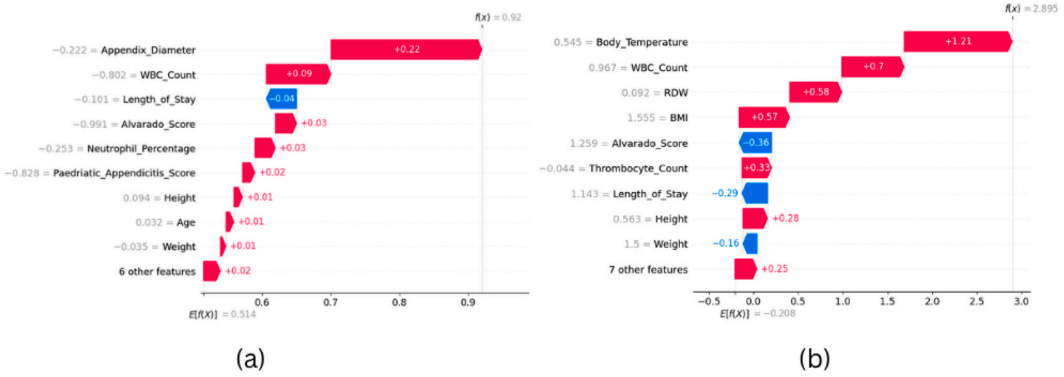


Fig. 9. Waterfall curve of (a) presumptive diagnosis, (b) peritonitis.

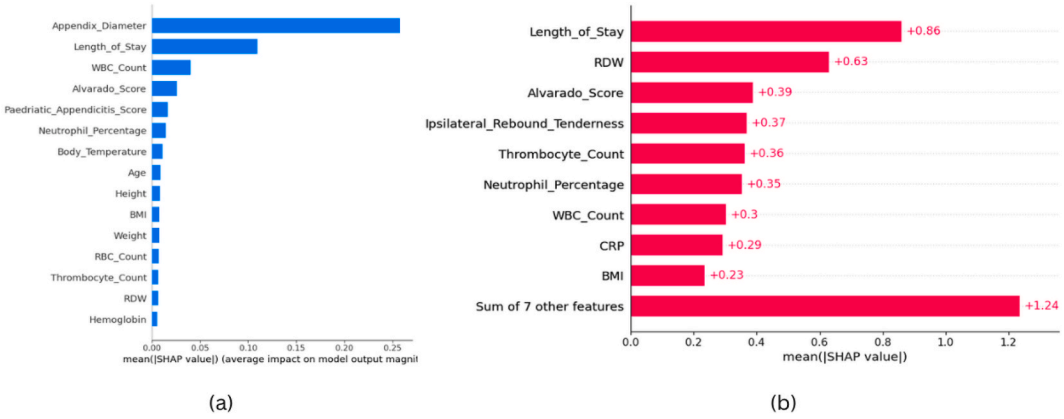


Fig. 10. Mean bar plot of (a) Presumptive Diagnosis, (b) Peritonitis.

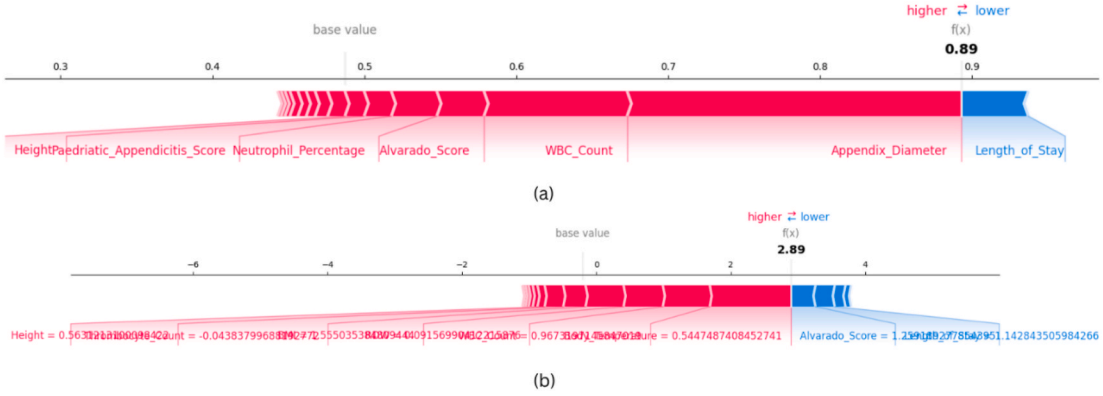


Fig. 11. Force plot of optimal model (a) Presumptive Diagnosis, (b) Peritonitis.

of peritonitis severity and the presumptive diagnosis of appendicitis. In the binary classification task of presumptive diagnosis of appendicitis, we found that children with a larger appendix diameter and higher WBC count were significantly more likely to be affected by appendicitis than those with a lower appendix diameter and WBC count. This highlights their importance in early presumptive detection—especially in paediatric cases where symptoms are often nonspecific. In contrast, the multiclass classification of peritonitis severity found that parameters such as Alvarado score and RDW were substantial predictors of disease progression, which is consistent with earlier research associating elevated RDW to systemic inflammation and more serious infections. Furthermore, thrombocyte count and ipsilateral rebound tenderness were found to be significantly linked with advanced peritoneal involvement. Other research on medical appendicitis revealed similar variations [42]. To ensure transparency and clinical interpretability, we deployed explainable AI tools like LIME, SHAP and Anchor, which confirmed the value of these attributes and provided case-specific insights, allowing for more informed judgments in paediatric care.

Marcinkevics et al. [8] and Akmese et al. [9] examined multiple models and reported similar good performance, with AUCs around 0.96, despite the lack of interpretability approaches. In this study, we used several predictive algorithms to forecast presumptive diagnosis and peritonitis. For presumptive diagnosis, the models had an AUC of 0.978 and an accuracy of 93.6%, while for peritonitis prediction, the AUC was 0.97 and the accuracy was 88.17%. Our work is distinguished by the employment of seven different XAI approaches, which contribute to

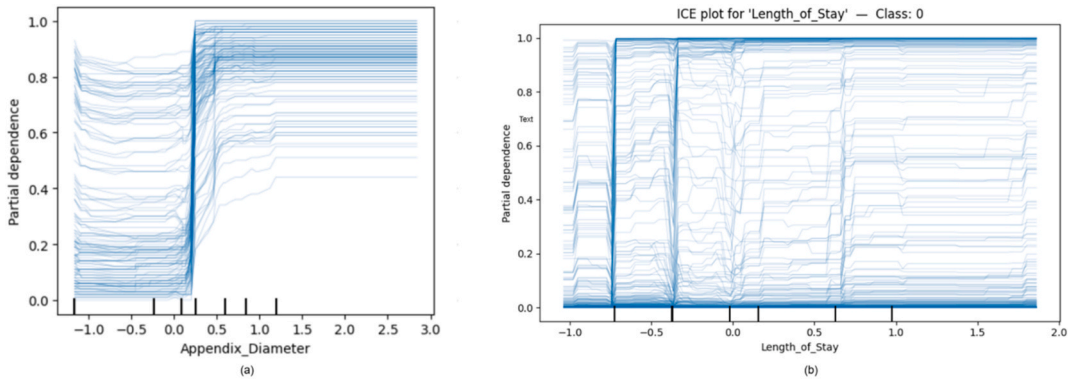


Fig. 12. ICE plot of (a) Presumptive Diagnosis, (b) Peritonitis.

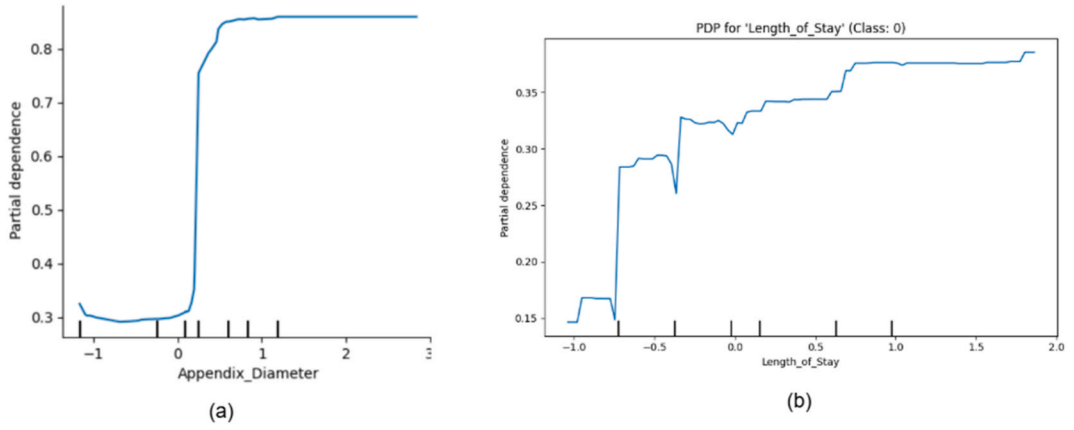


Fig. 13. PDP plot of (a) Presumptive Diagnosis, (b) Peritonitis.

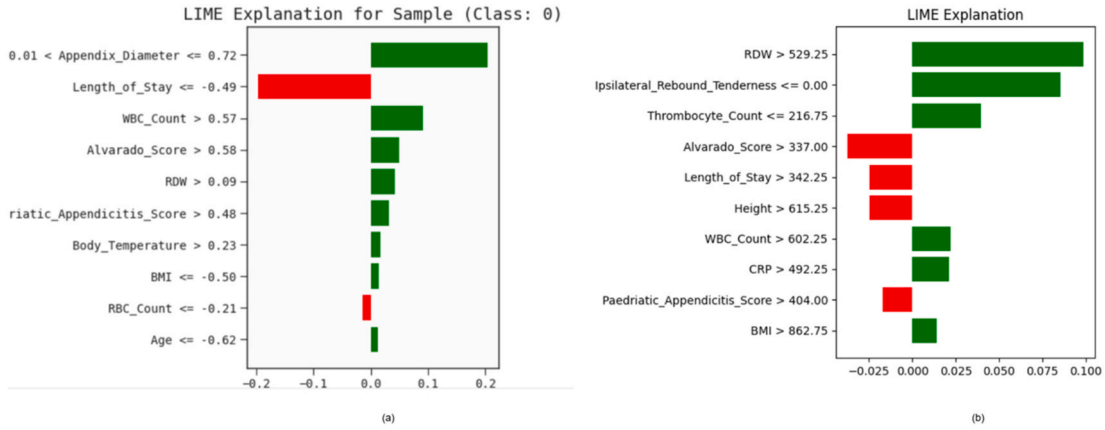


Fig. 14. LIME plot of (a) Presumptive Diagnosis, (b) Peritonitis.

more intelligible and therapeutically useful predictions. This comparative analysis is summarized in Table 10.

In addition to predicted performance and model interpretability, the actual implementation of AI-based clinical decision support systems necessitates secure patient information management. To address this deployment-related concern, an additional security assessment of the paediatric clinical dataset was performed using ChaCha20-Poly1305 authenticated encryption. The findings indicated that confidentiality and integrity may be maintained with a minimum computational and storage overhead, thus supporting the viability of secure clinical record storage and transmission in healthcare settings. Although the encryption procedure was not directly integrated into model training or inference, it adds to the proposed framework by addressing a critical requirement for real-world clinical adoption: the protection of sensitive patient data during storage, exchange, and system-wide deployment.

Even though this model fails to achieve excellent diagnosis accuracy, it acknowledges the AI gap between retrospective performance and clinical implementation. This issue was addressed by incorporating uncertainty quantification, which allows the algorithm to flag low-confidence predictions for radiologist evaluation [40]. When making critical medical decisions, this becomes an essential safety component. Recognizing the potential for gender-based bias, a subgroup analysis was conducted to ensure that performance is consistent across genders and age groups, thereby reducing common errors in medical computer

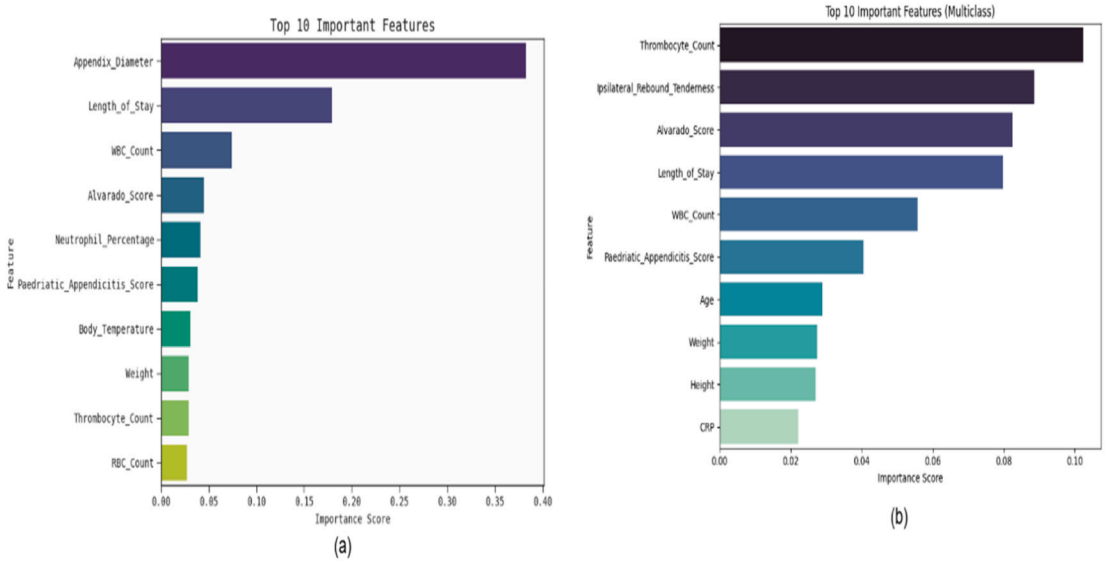


Fig. 15. Anchor plot of (a) Presumptive Diagnosis, (b) Peritonitis.

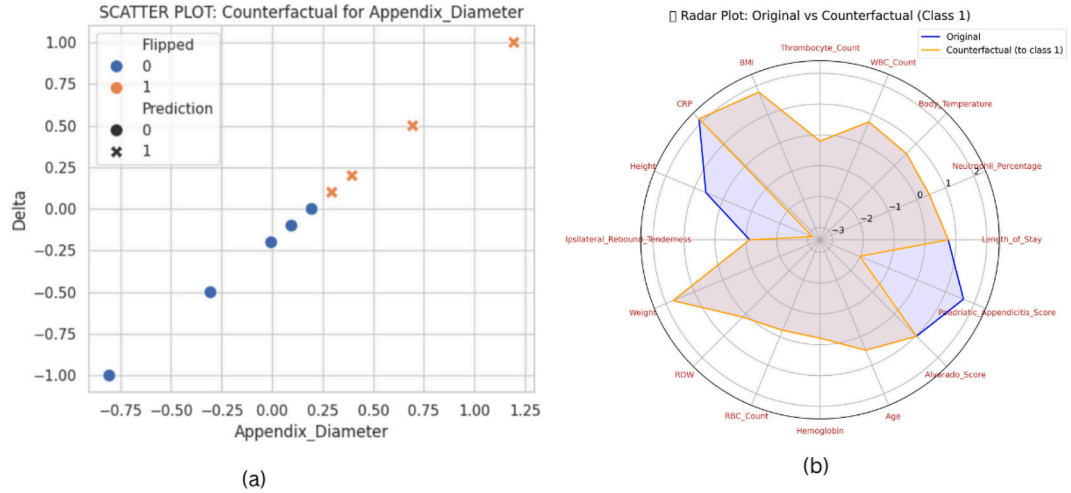


Fig. 16. Counterfactual explanation curve of (a) Presumptive Diagnosis, (b) Peritonitis.

vision. Furthermore, we acknowledge the lack of external validation at our single centre; nonetheless, we address this by emphasizing the likelihood of dataset shift when relocating to other clinical institutions and by providing 95% confidence intervals for all variables. To evaluate the robustness and generalizability of our findings across varied medical equipment and practice patterns, future research will focus on multi-centre evaluations.

This study has certain limitations. The dataset comprised 782 paediatric cases collected from a single hospital, which may limit the generalizability of the proposed models across different clinical settings, populations, and healthcare practices. Although rigorous cross-validation and explainable AI approaches were used, the lack of external validation prevents a thorough evaluation of the model's robustness and real-world application. Furthermore, the moderate dataset size limited the study of deep learning algorithms, which often demand larger and more diverse datasets for stable and consistent training.

These restrictions highlight the importance of collaborative, multi-institutional learning systems. In the future, this system can be enhanced to include Federated Learning (FL), which allows many hospitals to train predictive models jointly while keeping patient data local and private. By allowing institutions to learn from shared patterns without transferring raw clinical data, FL helps overcome important difficulties such as restricted data availability, poor generalizability, and privacy risks associated with centralized datasets [43]. Recent research integrating FL with Zero-Trust Architecture has yielded promising results for protecting sensitive medical information and preventing data leakage in distributed systems. Integrating these privacy-preserving and security-focused approaches would improve the proposed paediatric diagnostic framework's scalability, reliability, and real-world applicability across a wide range of clinical settings. Future research may potentially look into tighter integration of predictive modelling and security procedures using privacy-preserving machine learning paradigms like FL, secure inference, and encrypted distributed analytics.

5. Theoretical and practical implications

This study advances the application of interpretable machine-learning in pediatric emergency medicine by establishing a dual-target predictive framework.Traditional clinical scoring systems often operate as static rule-based approach or lack of granular probabilistic quantification; our findings demonstrate that combining advanced data

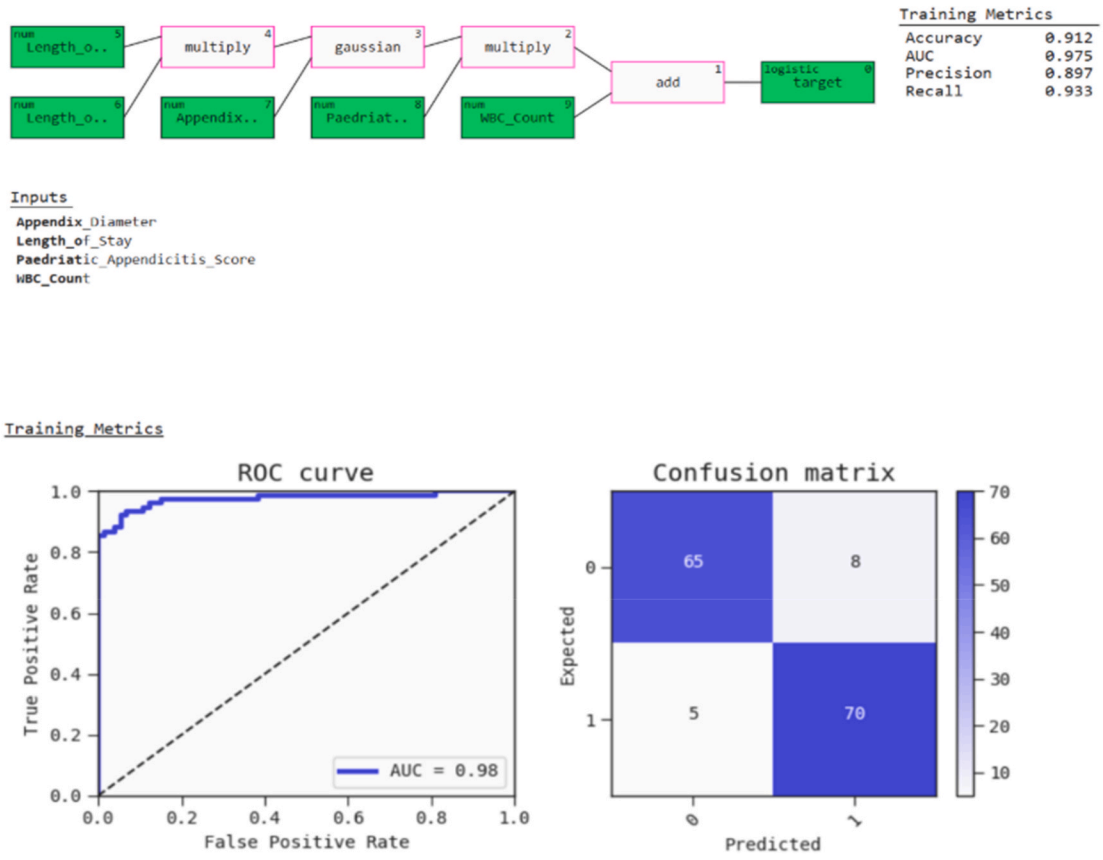


Fig. 17. Q-Lattice model output for binary classification of Presumptive Diagnosis.

Table 8
Important features influencing the prediction outcome based on XAI techniques.

Presumptive Diagnosis						
SHAP	LIME	Anchor	Q-Lattice	PDP	ICE	Counter Factual Explanation
Appendix diameter	Appendix diameter	Appendix diameter	Appendix diameter	Appendix diameter	Appendix diameter	Appendix diameter
Length of stay	Length of stay	Length of stay	Length of stay	Length of stay	Length of stay	Length of stay
WBC count	WBC count	WBC count	Paediatric appendicitis score	WBC count	WBC count	-
Alvarado score	Alvarado score	Alvarado score	WBC count	Alvarado score	Alvarado score	-
Paediatric appendicitis score	RDW	Neutrophil percentage	-	Neutrophil percentage	Neutrophil percentage	-
Neutrophil percentage	Paediatric appendicitis score	Paediatric appendicitis score	-	Paediatric appendicitis score	Paediatric appendicitis score	-
Body temperature	Body temperature	Body temperature	-	Body temperature	Body temperature	-

Peritonitis					
SHAP	LIME	Anchor	PDP	ICE	Counter Factual Explanation
Length of stay	RDW	Thrombocyte count	Length of stay	Length of stay	Paediatric appendicitis score
RDW	Ipsilateral rebound tenderness	Ipsilateral rebound tenderness	RDW	RDW	Height
Alvarado score	Alvarado score	Alvarado score	Thrombocyte count	Thrombocyte count	-
Ipsilateral rebound tenderness	Thrombocyte count	Length of stay	Alvarado score	Alvarado score	-
Thrombocyte count	Length of stay	WBC count	Ipsilateral rebound tenderness	Ipsilateral rebound tenderness	-
Neutrophil percentage	WBC count	Paediatric appendicitis score	Body temperature	Body temperature	-

resampling methods, such as Borderline-SMOTE and ADASYN, with gradient-boosted and voting classifiers successfully resolves severe class imbalance while preserving decision stability.Furthermore, integrating multi-model explainable artificial intelligence approach strengthens methodological transparency in healthcare analytics, as diverse attribution techniques consistently converge on core physiological markers

Table 9
Cryptographic evaluation results on paediatric dataset.

Metric	Value	Interpretation
Encryption time (ms)	2.178	Efficient encryption of structured dataset
Decryption time (ms)	2.806	Successful authenticated recovery
Storage overhead (%)	0.0	No ciphertext expansion
Diffusion	0.5007	Near-ideal avalanche effect
Confusion	0.9999	Near-maximum entropy
Integrity verification	Successful	Lossless data recovery



Fig. 18. Raw byte-level preview of the ChaCha20-Poly1305–encrypted paediatric dataset.

505fa7a8821339204be6687cd131b75f3381ed75f4aa1d84203ed
5a8b1c410ec90c4105d9952ed7c6c1529096008a7dd39b285f64c
4e2c0d27ac3c1f4a85eaa141b563b32f98a4f45af80da794a844a1
76ebedc4df39236ca2a1811da020d4a1c56681a59bb4b8f43ed23
40518f83c58531782c8f4ea0e565ec73aae4eb88033b4745fa40e6
cc172de57231891e70e875de982e54944b5790a4218bd78de088
e109bc5cd08d3695f39c43dd4b999f75582e7e86b71cb675b2113
279238db92adc92798100ead87e9f634973b7ef1bb044faa4cf456
b057010a37290b061d4cbca1eb32f2533e5ec3bd53700ffe7f8109
e68fbad760b694befc183fb1dac6db85

Fig. 19. Hexadecimal snapshot of the first 256 bytes of the encrypted paediatric dataset.

Table 10
Comparative analysis of studies that used AI to predict appendicitis.

Study Reference	Algorithms Applied	Optimal Scores	Interpretability Method
Marcinkevics et al. [8]	Logistic Regression, GBM, Random Forest	AUC- 0.96	None
Akmese et al. [9]	Neural Network, CART, Random Forest, Gradient Boosting, KNN, Logistic regression, SVM	Accuracy-96%	None
Our findings: I. Presumptive diagnosis II. Peritonitis	Logistic Regression, SVM, XGBoost, Gradient Boosting, Decision tree, KNN, AdaBoost, CatBoost	I. Accuracy-93.24% AUC-0.99 II. Accuracy-88.17% AUC-0.97	SHAP, LIME, Anchor, PDP, ICE, Counterfactual Explanation

such as appendix diameter and white blood cell counts.

From a practical perspective, the proposed framework offers actionable clinical support for clinicians, managing pediatric emergency cases where diagnostic ambiguity is high.The model provides a safety buffer that aids radiologists and emergency physicians in mitigating misdiagnosis risks by flagging low confidence predictions and incorporating uncertainty quantification. This identification of key operational and laboratory predictors helps the hospital make informed decisions.It helps the hospitals to refine protocols and allocate diagnostic imaging resources more efficiently.Additionally, the demonstrated feasibility of ChaCha20-Poly1305 authenticated encryption indicates that such clinical diagnostic systems can maintain data privacy and transmission security with minimal computational overhead, supporting real world deployment in distributed healthcare settings.

6. Conclusion

Explainable artificial intelligence approaches were used in this work to understand the predictions generated by different supervised machine learning models for paediatric appendicitis and peritonitis. The dataset was thoroughly pre-processed and statistically analyzed to identify relevant patterns and correct class imbalance with advanced oversampling methods such as Borderline-SMOTE, SVM-SMOTE, SMOTE, and ADASYN. Eight machine learning methods were used, and their effectiveness was measured across both diagnostic categories. For presumptive diagnosis, the Borderline-SMOTE and Soft voting combination produced the best results, with an accuracy of 93.24% and an AUC of 0.99. In the case of peritonitis prediction, the best results were obtained utilizing ADASYN and the XGBoost model, with an accuracy of 88.17% and an AUC of 0.97. To improve interpretability, seven XAI approaches (SHAP, LIME, Anchor, Q-Lattice, PDP, ICE, and Counterfactual Explanation) were used to emphasize the most influential aspects. These explainers identified appendix diameter, WBC count, Alvarado score, RDW, and length of stay as significant predictors. These findings support the use of interpretable machine learning as a useful tool for clinical decision-making in emergency care. In addition, a cryptographic evaluation was performed using the ChaCha20-Poly1305 encryption scheme, proving that structured paediatric clinical datasets may be safeguarded with minimal computational and storage expense.

Author contributions

Dyuthishree: Writing - Original Draft, Software; **Bhargavi Shenoy**: Writing - Original Draft, Software; **Niranjana Sampathila**: Methodology, Project administration; **Ujwal P**: Validation; **Krishnaraj Chadaga**: Conceptualization, Supervision. **Omkar Prabhu**: Conceptualization, Writing - Original Draft, Investigation, Formal Analysis, Methodology.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used generative AI tools for language refinement, editorial assistance, and manuscript-organization support. After using these tools, the authors critically reviewed, verified, and revised the content as needed. The authors take full responsibility for the accuracy, integrity, originality, and final content of the published article.

Funding

No Funding was obtained for this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank the Manipal Institute of Social Sciences, Humanities and Arts (MISHA), Manipal Academy of Higher Education, for their support in editing, proofreading, and formatting this manuscript, including grammatical corrections and overall language refinement.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.health.2026.100483.

Data availability

The data supporting the findings of this study include external data. The primary dataset used in this study is openly available in Zenodo Repository. https://zenodo.org/records/7669442.

References

[1] E. Aydın, et al., Diagnostic accuracy of a machine learning-derived appendicitis score in children: a multicenter validation study, Children 12 (7) (2025 Jul 16) 937.
[2] K. Chadaga, V. Khanna, S. Prabhu, N. Sampathila, R. Chadaga, S. Umakanth, D. Bhat, K.S. Swathi, R. Kamath, An interpretable and transparent machine learning framework for appendicitis detection in pediatric patients, Sci. Rep. 14 (1) (2024 Oct 18) 24454.
[3] Z. Kucukakcali, S. Akbulut, C. Colak, Evaluating ensemble-based machine learning models for diagnosing pediatric acute appendicitis: insights from a retrospective observational study, J. Clin. Med. 14 (12) (2025 Jun 16) 4264.
[4] M. Chekmeyan, S.H. Liu, Artificial intelligence for the diagnosis of pediatric appendicitis: a systematic review, Am. J. Emerg. Med. 92 (2025 Jun 1) 18–31.
[5] G.M. Alam, T. Biswas, S.A. Tanim, M.F. Mridha, An explainable analytics framework for predicting diabetes in women using Convolutional Neural Networks, Healthc. Anal. 10 (2025 Oct) 100422.
[6] K. Pang, A comparative study of explainable machine learning models with Shapley values for diabetes prediction, Healthc. Anal. 7 (2025 Jun 1) 100390.
[7] R. Ahmed, et al., A novel integrated logistic regression model enhanced with recursive feature elimination and explainable artificial intelligence for dementia prediction, Healthc. Anal. 6 (2024) 100362.
[8] R. Marcinkevics, P. Reis Wolfertstetter, S. Wellmann, C. Knorr, J.E. Vogt, Using machine learning to predict the diagnosis, management and severity of pediatric appendicitis, Front. Pediatr. 9 (2021 Apr 29) 662183.
[9] O.F. Akmese, G. Dogan, H. Kor, H. Erbay, E. Demir, The use of machine learning approaches for the diagnosis of acute appendicitis, Emerg. Med. Int. 2020 (1) (2020) 7306435.
[10] S. Bharati, M.R. Mondal, P. Podder, A review on explainable artificial intelligence for healthcare: why, how, and when? IEEE Trans. Artif. Intell. 5 (4) (2023) 1429–1442.
[11] J. Camacho-Cruz, P.O. Padilla, D.G. Sánchez, D.M. Mongui, L.K. Hoyos, M. L. Porras, P.V. Hoyos, F. Polo, C.A. Castro, Outcomes of acute appendicitis in patients younger than age 4: a descriptive study, Ann. of Pediatr. Sur. 18 (1) (2022) 59.
[12] V. Hassija, V. Chamola, A. Mahapatra, A. Singal, D. Goel, K. Huang, S. Scardapane, I. Spinelli, M. Mahmud, A. Hussain, Interpreting black-box models: a review on explainable artificial intelligence, Cogn. Comput. 16 (1) (2024 Jan) 45–74.
[13] S. Asadi, P. Jimeno-Sáez, A. López-Ballesteros, J. Senent-Aparicio, Comparison and integration of physical and interpretable AI-driven models for rainfall-runoff simulation, Results Eng. 24 (2024 Dec 1) 103048.
[14] L. Quakulinski, A. Koumpis, O.D. Beyan, Transparency in medical artificial intelligence systems, Int. J. Semantic Comput. (IJSC) 17 (4) (2023 Dec 19) 495–510.
[15] C. Rudin, Why black box machine learning should be avoided for high-stakes decisions, in brief, Nat. Rev. Methods Primers 2 (1) (2022 Oct 27) 81.
[16] S.Y. Chng, M.J. Tern, Y.S. Lee, L.T. Cheng, J. Kapur, J.G. Eriksson, Y.S. Chong, J. Savulescu, Ethical considerations in AI for child health and recommendations for child-centered medical AI, npj Digit. Med. 8 (1) (2025 Mar 10) 152.
[17] C. Combi, B. Amico, R. Bellazzi, A. Holzinger, J.H. Moore, M. Zitnik, J.H. Holmes, A manifesto on explainability for artificial intelligence in medicine, Artif. Intell. Med. 133 (2022 Nov 1) 102423.
[18] Y. Singh, Q.A. Hathaway, V. Keishing, S. Salehi, Y. Wei, N. Horvat, D.V. Vera-Garcia, A. Choudhary, Kh A. Mula, E. Quaia, J.B. Andersen, Beyond post hoc explanations: a comprehensive framework for accountable AI in medical imaging through transparency, interpretability, and explainability, Bioengineering 12 (8) (2025 Aug 15) 879.
[19] S. Gaube, H. Suresh, M. Raue, A. Merritt, S.J. Berkowitz, E. Lermer, J.F. Coughlin, J.V. Guttag, E. Colak, M. Ghassemi, Do as AI say: susceptibility in deployment of clinical decision-aids, npj Digit. Med. 4 (1) (2021 Feb 19) 31.
[20] R. Marcinkevičs, P. Reis Wolfertstetter, U. Klimiene, K. Chin-Cheong, A. Paschke, J. Zerres, M. Denzinger, D. Niederberger, S. Wellmann, E. Ozkan, C. Knorr, J. E. Vogt, Regensburg Pediatric Appendicitis Dataset (Version 1.03), Zenodo (2023), https://doi.org/10.5281/zenodo.7711412.
[21] A. Traverso, L. Wee, A. Dekker, R. Gillies, Repeatability and reproducibility of radiomic features: a systematic review, Int. J. Radiat. Oncol. Biol. Phys. 102 (4) (2018 Nov 15) 1143–1158.
[22] N. Pudjihartono, T. Fadason, A.W. Kempa-Liehr, J.M. O'Sullivan, A review of feature selection methods for machine learning-based disease risk prediction, Frontiers in bioinformatics 2 (2022 Jun 27) 927312.
[23] Z. Noroozi, A. Orooji, L. Erfannia, Analyzing the impact of feature selection methods on machine learning algorithms for heart disease prediction, Sci. Rep. 13 (1) (2023 Dec 18) 22588.
[24] S.K. Ghosh, A.H. Khandoker, Investigation on explainable machine learning models to predict chronic kidney diseases, Sci. Rep. 14 (1) (2024 Feb 14) 3687.
[25] W. Chen, K. Yang, Z. Yu, Y. Shi, C.P. Chen, A survey on imbalanced learning: latest research, applications and future directions, Artif. Intell. Rev. 57 (6) (2024 May 9) 137.
[26] A. Fernández, S. Garcia, F. Herrera, N.V. Chawla, SMOTE for learning from imbalanced data: progress and challenges, marking the 15-year anniversary, J. Artif. Intell. Res. 61 (2018 Apr 20) 863–905.
[27] N. Safaei, B. Safaei, S. Seyedekrami, M. Talafidaryani, A. Masoud, S. Wang, Q. Li, M. Moqri, E-CatBoost: an efficient machine learning framework for predicting ICU mortality using the eICU Collaborative Research Database, PLoS One 17 (5) (2022 May 5) e0262895.
[28] S. Sen, D. Kundu, K. Das, Variable selection for categorical response: a comparative study, Comput. Stat. 38 (2) (2023 Jun) 809–826.
[29] A. Pfob, S.C. Lu, C. Sidey-Gibbons, Machine learning in medicine: a practical introduction to techniques for data pre-processing, hyperparameter tuning, and model comparison, BMC Med. Res. Methodol. 22 (1) (2022 Nov 1) 282.
[30] J. Prithula, M.E. Chowdhury, M.S. Khan, K. Al-Ansari, S.M. Zughaier, K.R. Islam, A. Alqahtani, Improved pediatric ICU mortality prediction for respiratory diseases: machine learning and data subdivision insights, Respir. Res. 25 (1) (2024 May 23) 216.
[31] T.J. Bradshaw, Z. Huemann, J. Hu, A. Rahmim, A guide to cross-validation for artificial intelligence in medical imaging, Radiol. Artif. Intell. 5 (4) (2023 May 24) e220232.
[32] D. Wilimitis, C.G. Walsh, Practical considerations and applied examples of cross-validation for model development and evaluation in health care: tutorial, Jmir ai 2 (2023 Dec 18) e49023.
[33] A.B. Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. García, S. Gil-López, D. Molina, R. Benjamins, R. Chatila, Explainable Artificial Intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI, Inf. Fusion 58 (2020 Jun 1) 82–115.
[34] R. Hassan, et al., Unlocking the black box: enhancing human-AI collaboration in high-stakes healthcare scenarios through explainable AI, Technol. Forecast. Soc. Change 219 (2025) 124265.
[35] Y.L. Chou, C. Moreira, P. Bruza, C. Ouyang, J. Jorge, Counterfactuals and causability in explainable artificial intelligence: theory, algorithms, and applications, Inf. Fusion 81 (2022 May 1) 59–83.
[36] N. de Lacy, M. Ramshaw, W.Y. Lam, RiskPath: explainable deep learning for multistep biomedical prediction in longitudinal data, Patterns 6 (8) (2025 Aug 8).
[37] X. Li, P. Wang, Y. Zhu, W. Zhao, H. Pan, D. Wang, Interpretable machine learning model for predicting acute kidney injury in critically ill patients, BMC Med. Inf. Decis. Making 24 (1) (2024 May 31) 148.
[38] D. Bhati, F. Neha, M. Amiruzzaman, A survey on explainable artificial intelligence (XAI) techniques for visualizing deep learning models in medical imaging, J. Imaging 10 (10) (2024 Sep 25) 239.
[39] M.A. Lago, G. Zamzmi, B. Eich, J.G. Delfino, Evaluating explainability: a framework for systematic assessment of explainable AI features in medical imaging, Bioengineering 13 (1) (2026 Jan 16) 111.
[40] W. Jin, X. Li, M. Fatehi, G. Hamarneh, Guidelines and evaluation of clinical explainable AI in medical image analysis, Med. Image Anal. 84 (2023 Feb 1) 102684.
[41] A. Chaddad, J. Peng, J. Xu, A. Bouridane, Survey of explainable AI techniques in healthcare, Sensors 23 (2) (2023 Jan 5) 634.
[42] M. Issaiy, D. Zarei, A. Saghazadeh, Artificial intelligence and acute appendicitis: a systematic review of diagnostic and prognostic models, World J. Emerg. Surg. 18 (1) (2023 Dec 19) 59.
[43] S.H. Alsamhi, R. Myrzashova, A. Hawbani, S. Kumar, S. Srivastava, L. Zhao, X. Wei, M. Guizan, E. Curry, Federated learning meets blockchain in decentralized data sharing: Healthcare use case, IEEE Internet Things J. 11 (11) (2024 Feb 19) 19602–19615.